# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## MULTIPLEXER, DIGITAL, KEY CODE FRC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

## 2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

## 3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.26242 failures per million operating hours for class I or 0.61913 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FRC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/314-1 for class I and M28787/314-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.



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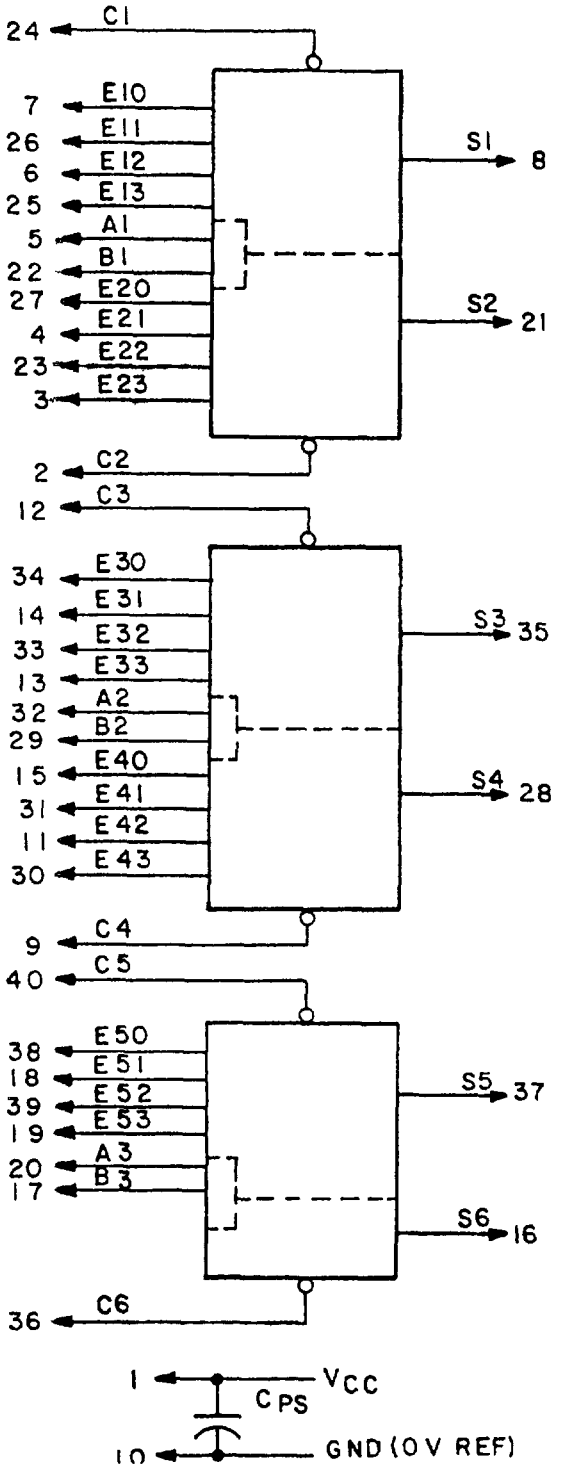



FIGURE 1. FRC module function diagram, multiplexer.



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TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation | | | | | | | | | | | | | |
| | Initial | III | S | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | MΩ |
| | End-of-life | III | | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | MΩ |
| B | $V_{CC}$ stress | IV | S | ← The module shall pass the remaining tests → | | | | | | | | | | |
| C1-C32 | Input breakdown current | V | S | 0 | 100 | 0 | 100 | 0 | 100 | 0 | 100 | 0 | 100 | µA |
| D1-D12 | Output leakage current, high level and low level | VI | X | -20 | 20 | -20 | 20 | -20 | 20 | -20 | 20 | -20 | 20 | µA |
| E1-E5 | Output short circuit current | VII | X | -20 | -130 | -20 | -130 | -20 | -130 | -20 | -130 | -20 | -130 | mA |
| F1-F64 | Functional | VIII | X | ← See table VIII → | | | | | | | | | | |
| G1-G32 | Input clamp diode voltage | IX | X | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | V |
| H1-H6 | Output voltage, low level | X | X | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| J1-J5 | Output voltage, high level | X | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| K1-K32 | Input current, high level | XI | X | -5 | 20 | -5 | 20 | -5 | 20 | -5 | 20 | -5 | 20 | µA |
| L1-L32 | Input current, low level | XII | X | 0 | -430 | 0 | -430 | 0 | -430 | 0 | -430 | 0 | -430 | µA |
| M1, M2 | Power supply current, high level and low level output | XIII | X | 15 | 36 | 15 | 36 | 15 | 36 | 15 | 36 | 15 | 36 | mA |
| M3 | Power supply current, high impedance output | XIII | X | 15 | 42 | 15 | 42 | 15 | 42 | 15 | 42 | 15 | 42 | mA |
| N1-N20 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 5 | 29 | 5 | 31 | 5 | 31 | 5 | 33 | 5 | 33 | ns |
| N21-N30 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 5 | 49 | 5 | 51 | 5 | 51 | 5 | 53 | 5 | 53 | ns |
| N31-N35 | Propagation delay time ($t_{PHZ}$), $C_L$ = 50 pF | XIV | S | 5 | 70 | 5 | 75 | 5 | 75 | 5 | 80 | 5 | 80 | ns |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| N36-N41 | Propagation delay time ($t_{PLZ}$), $C_L$ = 50 pF | XIV | S | 5 | 35 | 5 | 40 | 5 | 40 | 5 | 45 | 5 | 45 | ns |
| N42-N61 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 5 | 39 | 5 | 41 | 5 | 41 | 5 | 43 | 5 | 43 | ns |
| N62-N71 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 5 | 59 | 5 | 61 | 5 | 61 | 5 | 63 | 5 | 63 | ns |
| P1-P20 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 5 | 24 | 5 | 26 | 5 | 26 | 5 | 28 | 5 | 28 | ns |
| P21-P30 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 5 | 36 | 5 | 38 | 5 | 38 | 5 | 40 | 5 | 40 | ns |
| P31-P35 | Propagation delay time ($t_{PZH}$), $C_L$ = 50 pF | XIV | S | 5 | 50 | 5 | 55 | 5 | 55 | 5 | 60 | 5 | 60 | ns |
| P36-P41 | Propagation delay time ($t_{PZL}$), $C_L$ = 50 pF | XIV | S | 5 | 30 | 5 | 35 | 5 | 35 | 5 | 40 | 5 | 40 | ns |
| P42-P61 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 5 | 34 | 5 | 36 | 5 | 36 | 5 | 38 | 5 | 38 | ns |
| P62-P71 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 5 | 46 | 5 | 48 | 5 | 48 | 5 | 50 | 5 | 50 | ns |
| Q1-Q6 | Output crosstalk, low level | XV | Q | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| R1-R5 | Output crosstalk, high level | XV | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| S | Power supply transient amplitude | XVI | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| T | Power supply filter capacitance | XVII | S | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | µF |
| U1-U32 | Input capacitance | XVIII | Q | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | pF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.





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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If tests conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

| | Class I | Class II | Unit |
|---|---|---|---|
| Maximum power dissipation (output not loaded) | 231 | 231 | mW |



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| | Class I | Class II | Unit |
|---|---|---|---|
| Input current | | | |
| Low level ($V_{IL}$ = 0.4 V) | -430 max | -430 max | µA |
| High level ($V_{IH}$ = 2.4 V) | 20 max | 20 max | µA |
| Output drive current | | | |
| Low level ($V_{OL}$ = 0.4 V) | 4 max | 4 max | mA |
| High level ($V_{OH}$ = 2.4 V) | -1 max | -1 max | mA |
| Output leakage current | | | |
| Low level and high level | ±20 | ±20 | µA |

6.5 Module function. This module is a 3-state 4 to 1 multiplexer.

6.5.1 Input to output relationship. Each multiplexer has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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TABLE II. Input to output relationship. 1/ - 4/

| Inputs | | | | | | | Outputs |
|---|---|---|---|---|---|---|---|
| A1 | B1 | E10 | E11 | E12 | E13 | C1 | S1 |
| | | E20 | E21 | E22 | E23 | C2 | S2 |
| A2 | B2 | E30 | E31 | E32 | E33 | C3 | S3 |
| | | E40 | E41 | E42 | E43 | C4 | S4 |
| A3 | B3 | E50 | E51 | E52 | E53 | C5 | S5 |
| X | X | X | X | X | X | H | HZ |
| L | L | L | X | X | X | L | L |
| L | L | H | X | X | X | L | H |
| L | H | X | L | X | X | L | L |
| L | H | X | H | X | X | L | H |
| H | L | X | X | L | X | L | L |
| H | L | X | X | H | X | L | H |
| H | H | X | X | X | L | L | L |
| H | H | X | X | X | H | L | H |

| Inputs | | | Outputs |
|---|---|---|---|
| A3 | B3 | C6 | S6 |
| X | X | H | HZ |
| X | X | L | L |

1/ X = Indifferent level.

2/ L = Low level.

3/ H = High level.

4/ HZ = High impedance.



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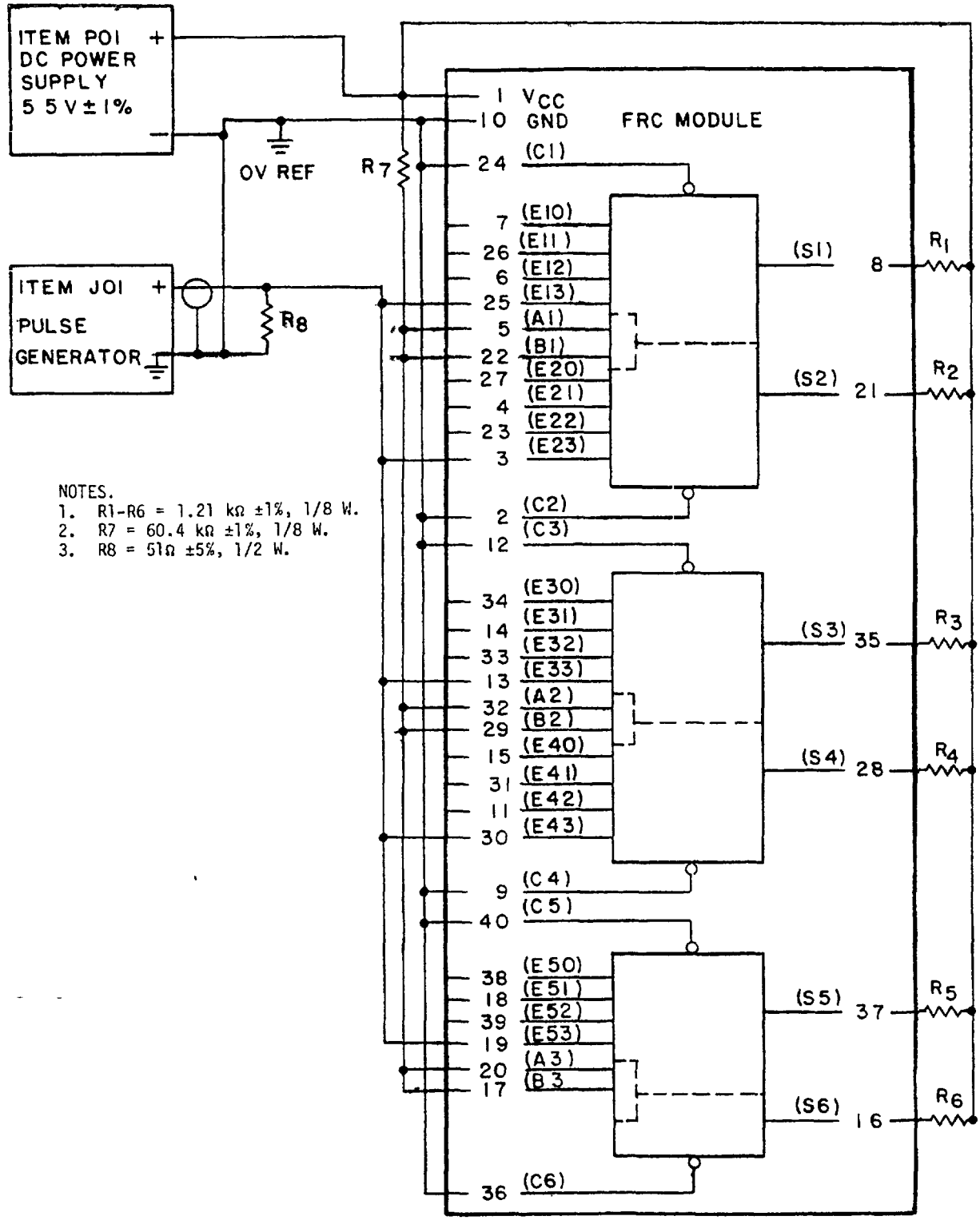
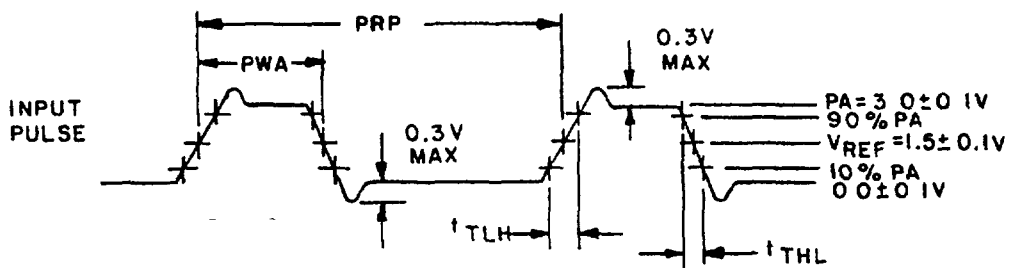



FIGURE 2. Operating temperature and life test circuit.



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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 6 \pm 2$ ns.
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.

FIGURE 3. Operating temperature and life test circuit waveform for figure 2.



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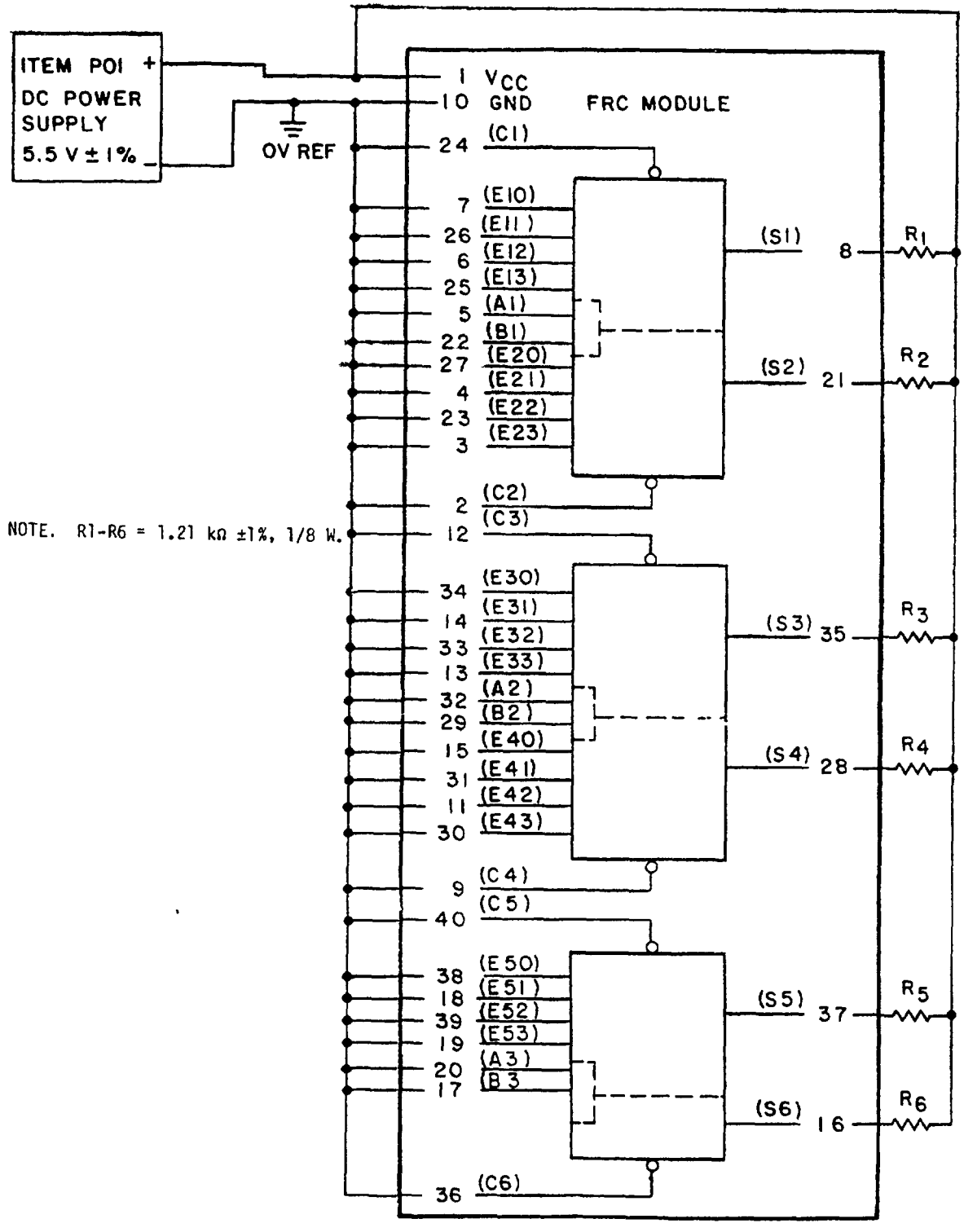



FIGURE 4. Component temperature test circuit



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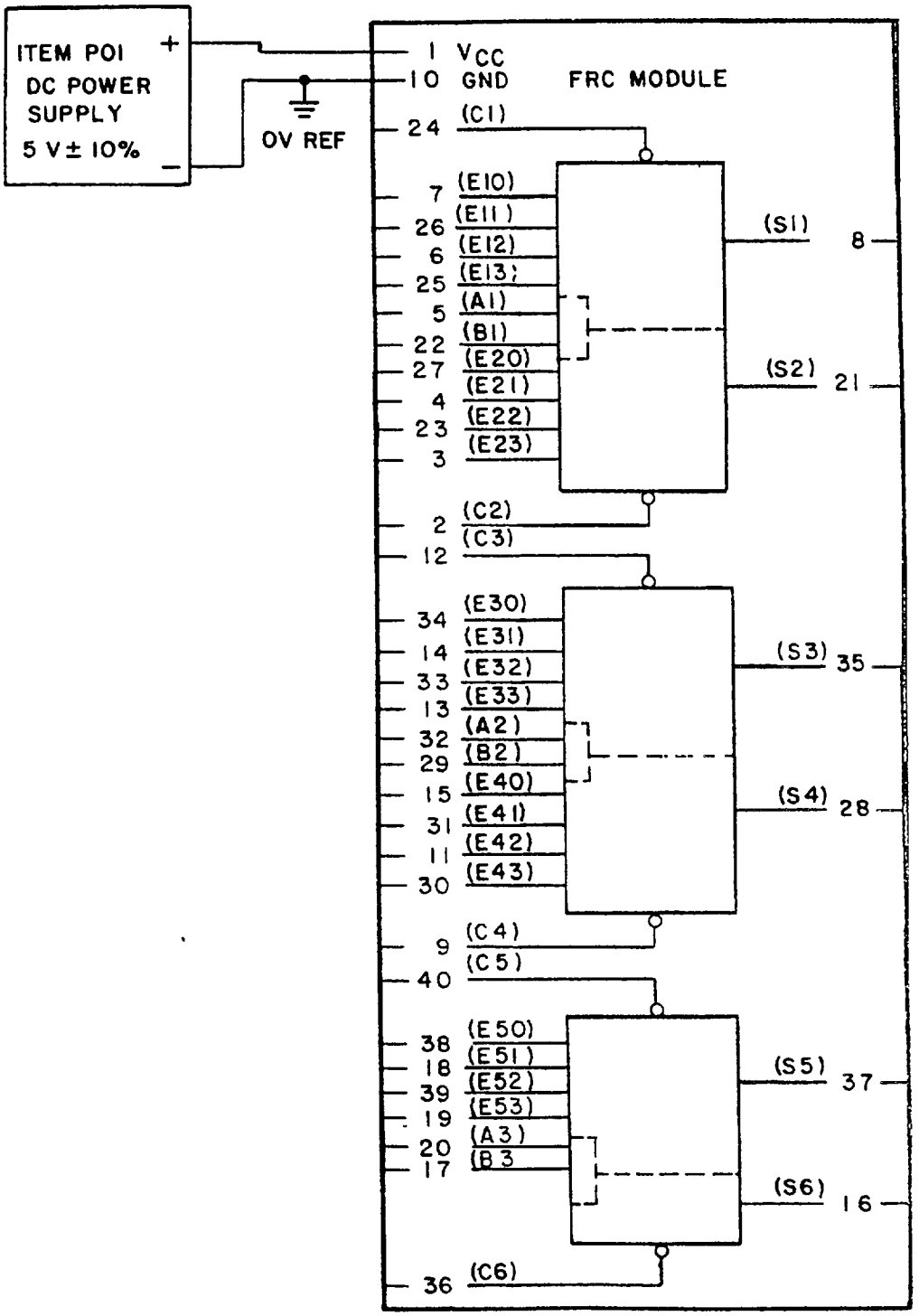



FIGURE 5. Vibration test circuit.



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TABLE III. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation | C | Frame | 10 | ∞ | MΩ | 6 |

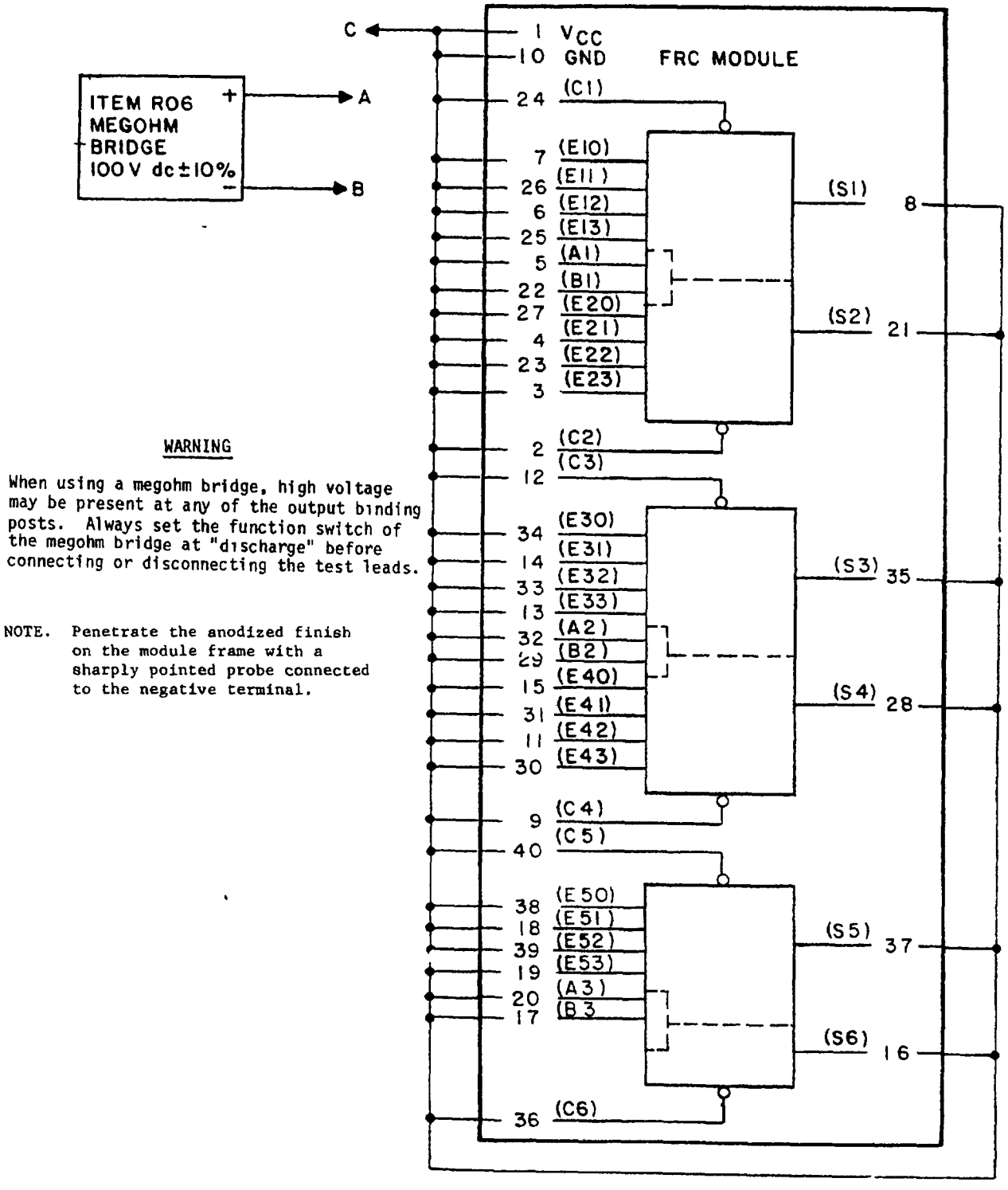
TABLE IV. $V_{CC}$ stress test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| B | $V_{CC}$ stress | The module shall pass the remaining tests. | | --- | 7 |





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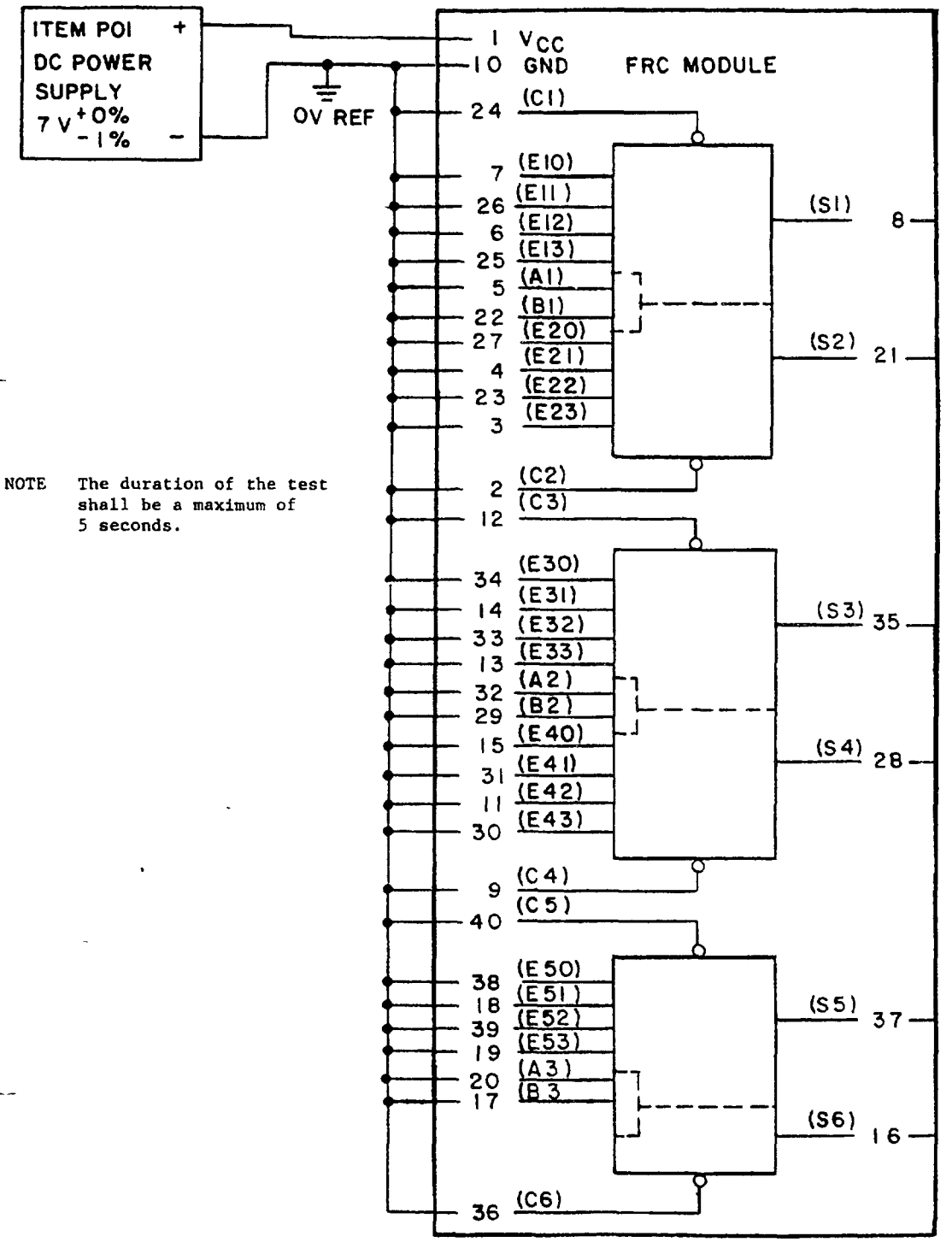
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.



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NOTE The duration of the test shall be a maximum of 5 seconds.

FIGURE 7 $V_{CC}$ stress test circuit for the table IV.



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TABLE V. Input breakdown current test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C1 | C1 Input breakdown current | Other inputs | 24 | 0 | 100 | µA | 8 |
| C2 | E10 " | " | 7 | " | " | " | " |
| C3 | E11 " | " | 26 | " | " | " | " |
| C4 | E12 " | " | 6 | " | " | " | " |
| C5 | E13 " | " | 25 | " | " | " | " |
| C6 | A1 " | " | 5 | " | " | " | " |
| C7 | B1 " | " | 22 | " | " | " | " |
| C8 | E20 " | " | 27 | " | " | " | " |
| C9 | E21 " | " | 4 | " | " | " | " |
| C10 | E22 " | " | 23 | " | " | " | " |
| C11 | E23 " | " | 3 | " | " | " | " |
| C12 | C2 " | " | 2 | " | " | " | " |
| C13 | C3 " | " | 12 | " | " | " | " |
| C14 | E30 " | " | 34 | " | " | " | " |
| C15 | E31 " | " | 14 | " | " | " | " |
| C16 | E32 " | " | 33 | " | " | " | " |
| C17 | E33 " | " | 13 | " | " | " | " |
| C18 | A2 " | " | 32 | " | " | " | " |
| C19 | B2 " | " | 29 | " | " | " | " |
| C20 | E40 " | " | 15 | " | " | " | " |
| C21 | E41 " | " | 31 | " | " | " | " |
| C22 | E42 " | " | 11 | " | " | " | " |
| C23 | E43 " | " | 30 | " | " | " | " |
| C24 | C4 " | " | 9 | " | " | " | " |
| C25 | C5 " | " | 40 | " | " | " | " |



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TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| C26 | E50 | Input breakdown current | Other inputs | 38 | 0 | 100 | µA | 8 |
| C27 | E51 | " | " | 18 | " | " | " | " |
| C28 | E52 | " | " | 39 | " | " | " | " |
| C29 | E53 | " | " | 19 | " | " | " | " |
| C30 | A3 | " | " | 20 | " | " | " | " |
| C31 | B3 | " | " | 17 | " | " | " | " |
| C32 | C6 | " | " | 36 | " | " | " | " |



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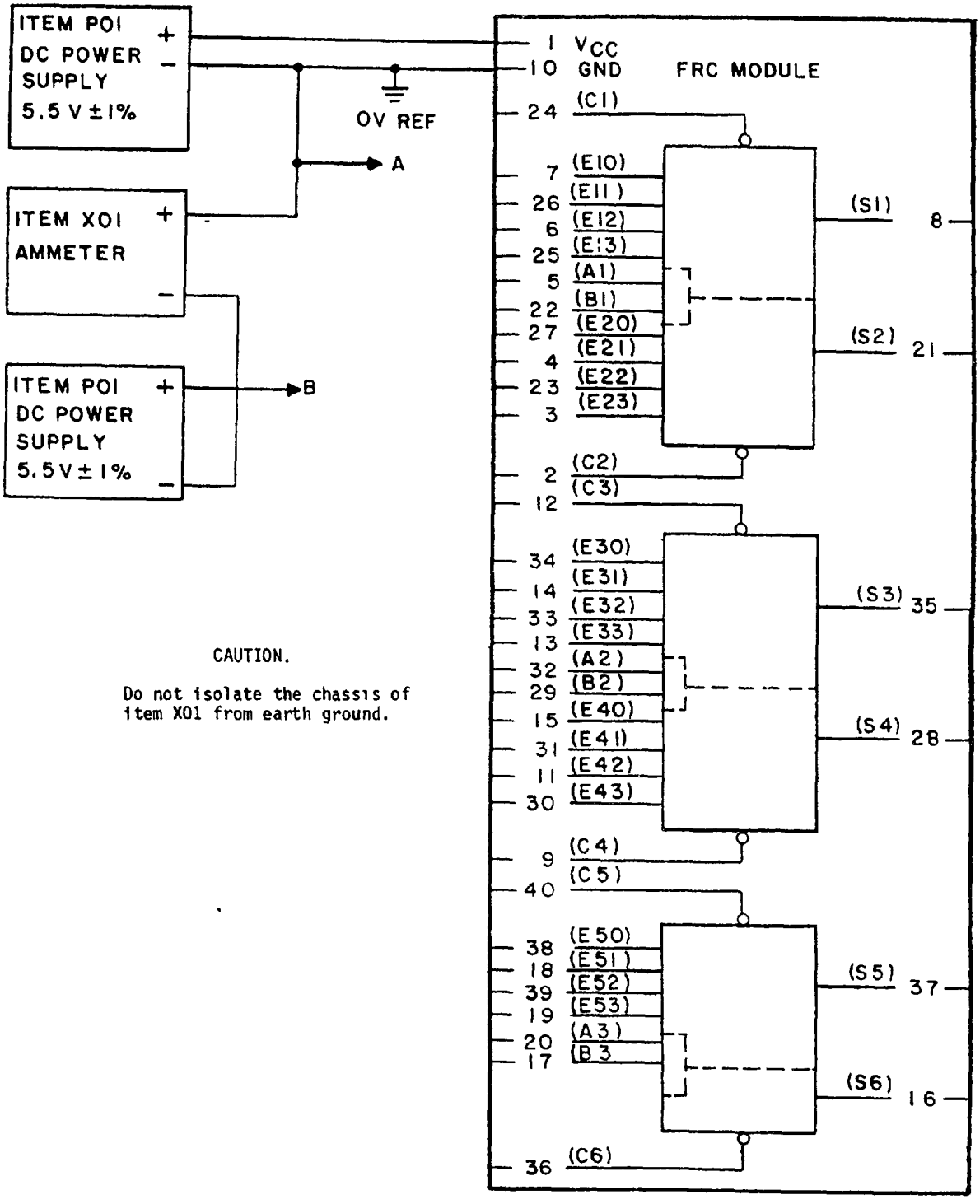



FIGURE 8. Input breakdown current test circuit for table V



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TABLE VI. Output leakage current (high-level and low-level) test.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| D1 | S1 | Output leakage current, high level | 8 | -20 | 20 | µA | 9 |
| D2 | S2 | " | 21 | " | " | " | " |
| D3 | S3 | " | 35 | " | " | " | " |
| D4 | S4 | " | 28 | " | " | " | " |
| D5 | S5 | " | 37 | " | " | " | " |
| D6 | S6 | " | 16 | " | " | " | " |
| D7 | S1 | Output leakage current, low level | 8 | " | " | " | " |
| D8 | S2 | " | 21 | " | " | " | " |
| D9 | S3 | " | 35 | " | " | " | " |
| D10 | S4 | " | 28 | " | " | " | " |
| D11 | S5 | " | 37 | " | " | " | " |
| D12 | S6 | " | 16 | " | " | " | " |

TABLE VII. Output short circuit current test.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| E1 | S1 | Output short circuit current | 8 | -20 | -130 | mA | 10 |
| E2 | S2 | " | 21 | " | " | " | " |
| E3 | S3 | " | 35 | " | " | " | " |
| E4 | S4 | " | 28 | " | " | " | " |
| E5 | S5 | " | 37 | " | " | " | " |



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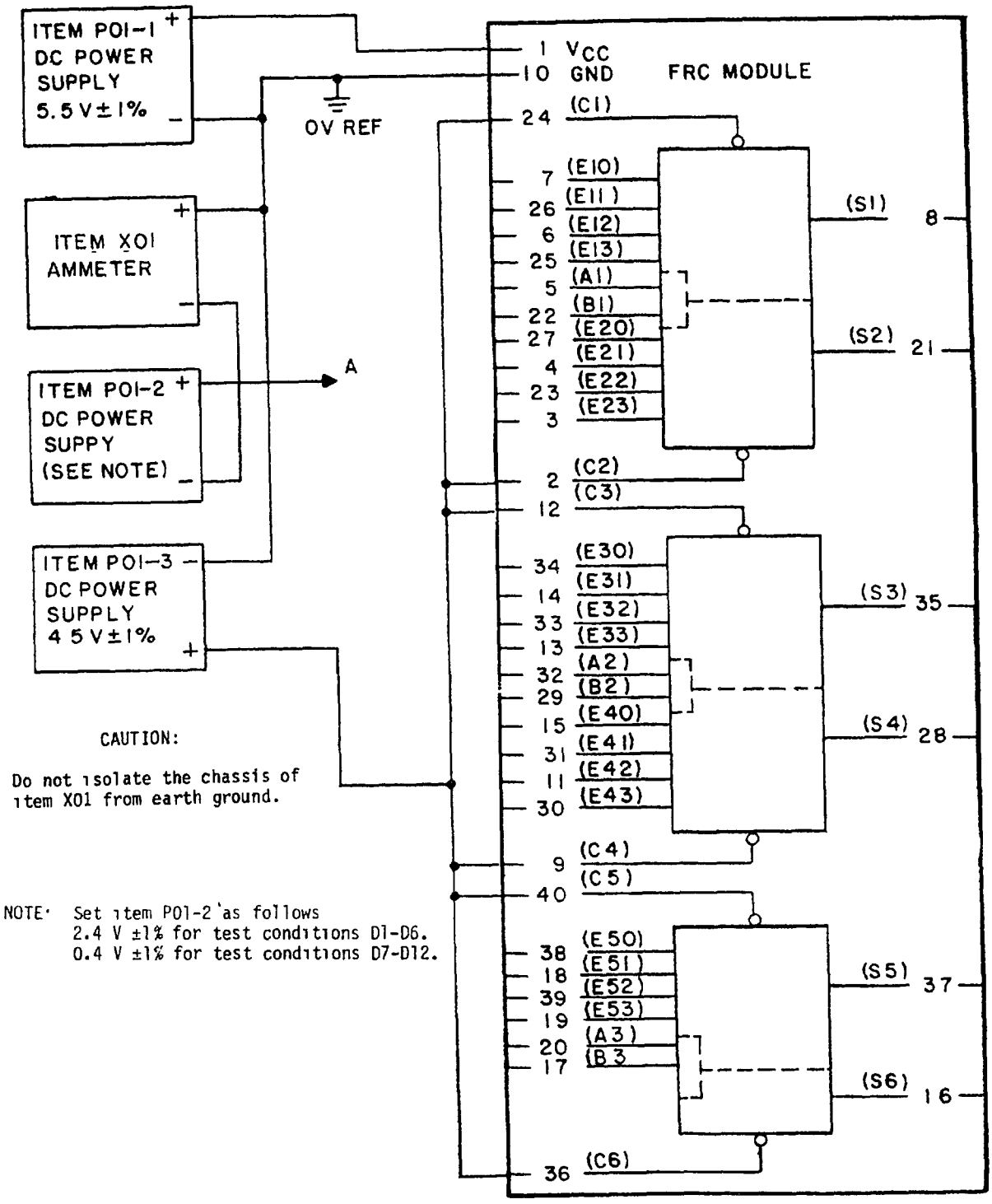



FIGURE 9. Output leakage current (high-level and low-level) test circuit for table VI.



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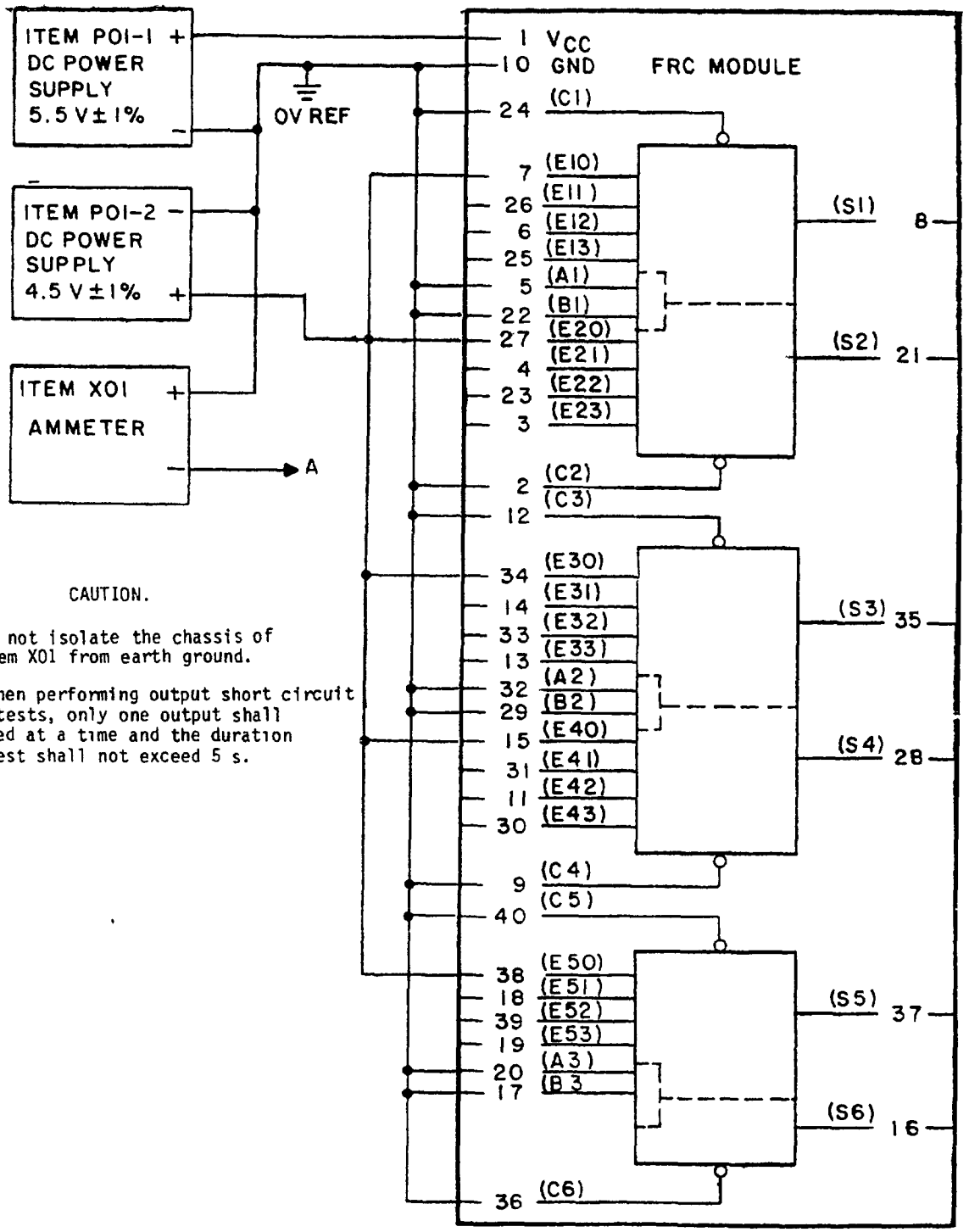



FIGURE 10. Output short circuit current test circuit for table VII.



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TABLE VIII. Functional test.

| Test condition | Parameter | Connect | | | Outputs | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | S1 | S2 | S3 | S4 | S5 | S6 | Min | Max | | |
| F1 | Functional | Open | E, F, G, H, K, L, D | 1/ | L | L | L | L | L | L | | 2/ | V | 11 |
| F2 | " | L | D, E, F, G, H, K | " | L | L | L | L | L | L | | " | " | " |
| F3 | " | K | E, F, G, H, L, D | " | L | L | L | L | L | L | | " | " | " |
| F4 | " | K, L | D, E, F, G, H | " | L | L | L | L | L | L | | " | " | " |
| F5 | " | E | F, G, H, D, K, L | " | H | H | H | H | H | L | | " | " | " |
| F6 | " | E, L | F, G, H, D, K | " | L | L | L | L | L | L | | " | " | " |
| F7 | " | E, K | F, G, H, D, L | " | L | L | L | L | L | L | | " | " | " |
| F8 | " | E, K, L | F, G, H, D | " | L | L | L | L | L | L | | " | " | " |
| F9 | " | F | E, G, H, D, K, L | " | L | L | L | L | L | L | | " | " | " |
| F10 | " | F, L | E, G, H, D, K | " | H | H | H | H | H | L | | " | " | " |
| F11 | " | F, K | E, G, H, D, L | " | L | L | L | L | L | L | | " | " | " |
| F12 | " | F, K, L | E, G, H, D | " | L | L | L | L | L | L | | " | " | " |
| F13 | " | E, F | G, H, D, K, L | " | H | H | H | H | H | L | | " | " | " |
| F14 | " | E, F, L | G, H, D, K | " | H | H | H | H | H | L | | " | " | " |
| F15 | " | E, F, K | G, H, D, L | " | L | L | L | L | L | L | | " | " | " |

See footnotes at end of table.



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TABLE VIII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Outputs | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | S1 | S2 | S3 | S4 | S5 | S6 | Min | Max | | |
| F16 | Functional | E, F, K, L | G, H, D | 1/ | L | L | L | L | L | L | | 2/ | V | 11 |
| F17 | " | G | E, F, H, D, K, L | " | L | L | L | L | L | L | | " | " | " |
| F18 | " | G, L | E, F, H, D, K | " | L | L | L | L | L | L | | " | " | " |
| F19 | " | G, K | E, F, H, D, L | " | H | H | H | H | H | L | | " | " | " |
| F20 | " | G, K, L | E, F, H, D | " | L | L | L | L | L | L | | " | " | " |
| F21 | " | G, E | F, H, D, K, L | " | H | H | H | H | H | L | | " | " | " |
| F22 | " | G, E, L | F, H, D, K | " | L | L | L | L | L | L | | " | " | " |
| F23 | " | G, E, K | F, H, D, L | " | H | H | H | H | H | L | | " | " | " |
| F24 | " | G, E, K, L | F, H, D | " | L | L | L | L | L | L | | " | " | " |
| F25 | " | G, F | E, H, D, K, L | " | L | L | L | L | L | L | | " | " | " |
| F26 | " | G, F, L | E, H, D, K | " | H | H | H | H | H | L | | " | " | " |
| F27 | " | G, F, K | E, H, D, L | " | H | H | H | H | H | L | | " | " | " |
| F28 | " | G, F, K, L | E, H, D | " | L | L | L | L | L | L | | " | " | " |
| F29 | " | G, F, E | H, D, K, L | " | H | H | H | H | H | L | | " | " | " |

See footnotes at end of table.



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TABLE VIII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Outputs | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | S1 | S2 | S3 | S4 | S5 | S6 | Min | Max | | |
| F30 | Functional | G, F, E, L | H, D, K | 1/ | H | H | H | H | H | L | 2/ | | V | 11 |
| F31 | " | G, F, E, K | H, D, L | " | H | H | H | H | H | L | " | | " | " |
| F32 | " | G, F, E, K, L | H, D | " | L | L | L | L | L | L | " | | " | " |
| F33 | " | H | E, F, G, D, K, L | " | L | L | L | L | L | L | " | | " | " |
| F34 | " | H, L | E, F, G, D, K | " | L | L | L | L | L | L | " | | " | " |
| F35 | " | H, K | E, F, G, D, L | " | L | L | L | L | L | L | " | | " | " |
| F36 | " | H, K, L | E, F, G, D | " | H | H | H | H | H | L | " | | " | " |
| F37 | " | H, E | F, G, D, K, L | " | H | H | H | H | H | L | " | | " | " |
| F38 | " | H, E, L | F, G, D, K | " | L | L | L | L | L | L | " | | " | " |
| F39 | " | H, E, K | F, G, D, L | " | L | L | L | L | L | L | " | | " | " |
| F40 | " | H, E, K, L | F, G, D | " | H | H | H | H | H | L | " | | " | " |
| F41 | " | H, F | E, G, D, K, L | " | L | L | L | L | L | L | " | | " | " |
| F42 | " | H, F, L | E, G, D, K | " | H | H | H | H | H | L | " | | " | " |
| F43 | " | H, F, K | E, G, D, L | " | L | L | L | L | L | L | " | | " | " |

See footnotes at end of table.





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TABLE VIII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Outputs | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | S1 | S2 | S3 | S4 | S5 | S6 | Min | Max | | |
| F44 | Functional | H, F, K, L | E, G, D | 1/ | H | H | H | H | H | L | | 2/ | V | 11 |
| F45 | " | H, F, E | G, D, K, L | " | H | H | H | H | H | L | | " | " | " |
| F46 | " | H, F, E, L | G, D, K | " | H | H | H | H | H | L | | " | " | " |
| F47 | " | H, F, E, K | G, D, L | " | L | L | L | L | L | L | | " | " | " |
| F48 | " | H, F, E, K, L | G, D | " | H | H | H | H | H | L | | " | " | " |
| F49 | " | H, G | F, E, D, K, L | " | L | L | L | L | L | L | | " | " | " |
| F50 | " | H, G, L | F, E, D, K | " | L | L | L | L | L | L | | " | " | " |
| F51 | " | H, G, K | F, E, D, L | " | H | H | H | H | H | L | | " | " | " |
| F52 | " | H, G, K, L | F, E, D | " | H | H | H | H | H | L | | " | " | " |
| F53 | " | H, G, E | F, D, K, L | " | H | H | H | H | H | L | | " | " | " |
| F54 | " | H, G, E, L | F, D, K | " | L | L | L | L | L | L | | " | " | " |
| F55 | " | H, G, E, K | F, D, L | " | H | H | H | H | H | L | | " | " | " |
| F56 | " | H, G, E, K, L | F, D | " | H | H | H | H | H | L | | " | " | " |
| F57 | " | H, G, F | E, D, K, L | " | L | L | L | L | L | L | | " | " | " |

See footnotes at end of table.



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TABLE VIII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Outputs | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | S1 | S2 | S3 | S4 | S5 | S6 | Min | Max | | |
| F58 | Functional | H, G, F, L | E, D, K | 1/ | H | H | H | H | H | L | 2/ | | V | 11 |
| F59 | " | H, G, F, K | E, D, L | " | H | H | H | H | H | L | " | | " | " |
| F60 | " | H, G, F, K, L | E, D | " | H | H | H | H | H | L | " | | " | " |
| F61 | " | H, G, F, E | D, K, L | " | H | H | H | H | H | L | " | | " | " |
| F62 | " | H, G, F, E, L | D, K | " | H | H | H | H | H | L | " | | " | " |
| F63 | " | H, G, F, E, K | D, L | " | H | H | H | H | H | L | " | | " | " |
| F64 | " | H, G, F, E, K, L | D | " | H | H | H | H | H | L | " | | " | " |

1/ For test conditions F1-F64, the output level shall be checked by successively connecting point C to each of the outputs (outputs S1, S2, S3, S4, S5, and S6 shall be checked on pins 8, 21, 35, 28, 37, and 16).

2/ Outputs:
L = low level (0.4 V maximum).
H = high level (2.4 V minimum).





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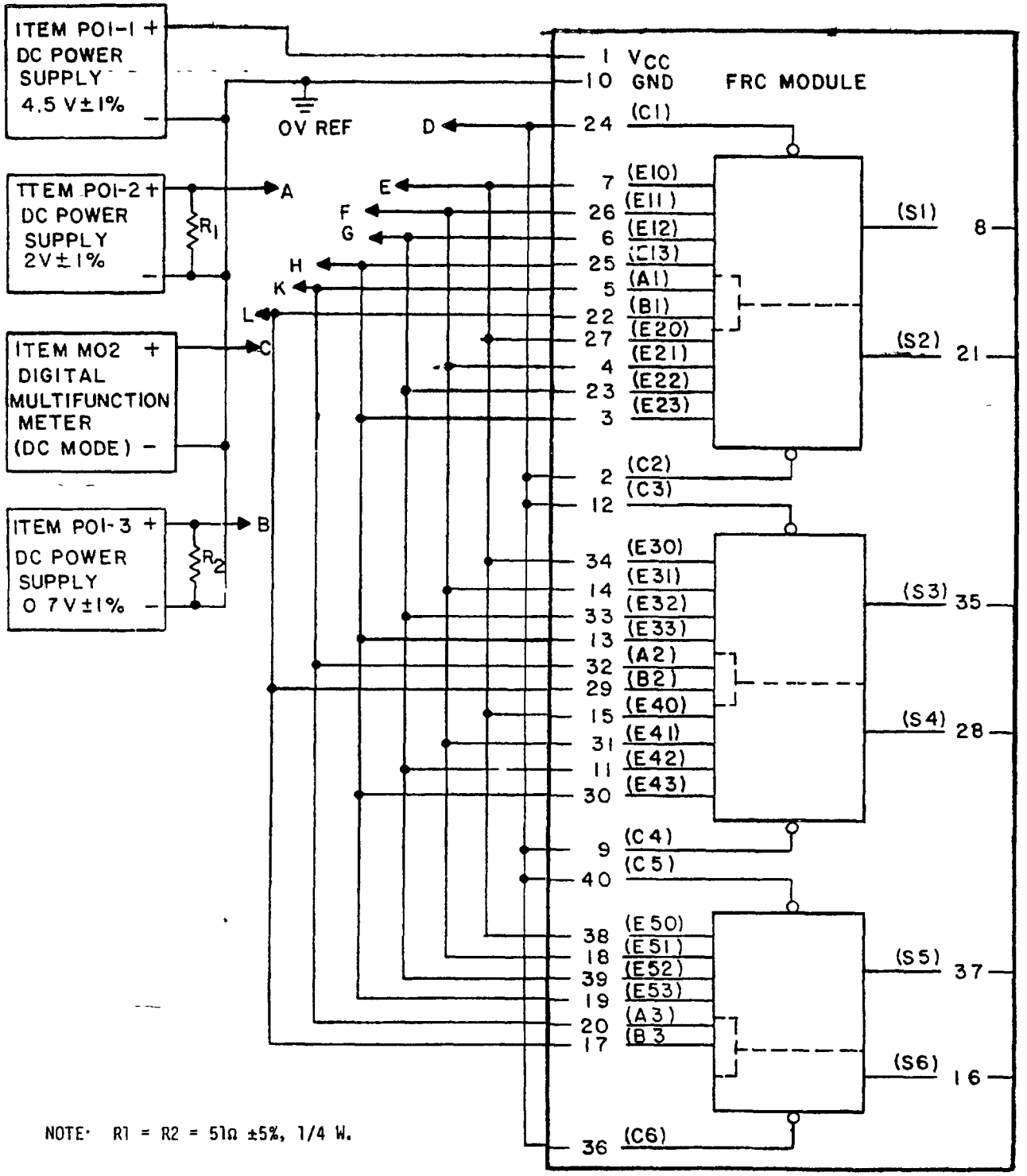



FIGURE 11. Functional test circuit for table VIII.



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TABLE IX. Input clamp diode voltage test.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| G1 | C1 | Input clamp diode voltage | 24 | -0.4 | -1.5 | V | 12 |
| G2 | E10 | " | 7 | " | " | " | " |
| G3 | E11 | " | 26 | " | " | " | " |
| G4 | E12 | " | 6 | " | " | " | " |
| G5 | E13 | " | 25 | " | " | " | " |
| G6 | A1 | " | 5 | " | " | " | " |
| G7 | B1 | " | 22 | " | " | " | " |
| G8 | E20 | " | 27 | " | " | " | " |
| G9 | E21 | " | 4 | " | " | " | " |
| G10 | E22 | " | 23 | " | " | " | " |
| G11 | E23 | " | 3 | " | " | " | " |
| G12 | C2 | " | 2 | " | " | " | " |
| G13 | C3 | " | 12 | " | " | " | " |
| G14 | E30 | " | 34 | " | " | " | " |
| G15 | E31 | " | 14 | " | " | " | " |
| G16 | E32 | " | 33 | " | " | " | " |
| G17 | E33 | " | 13 | " | " | " | " |
| G18 | A2 | " | 32 | " | " | " | " |





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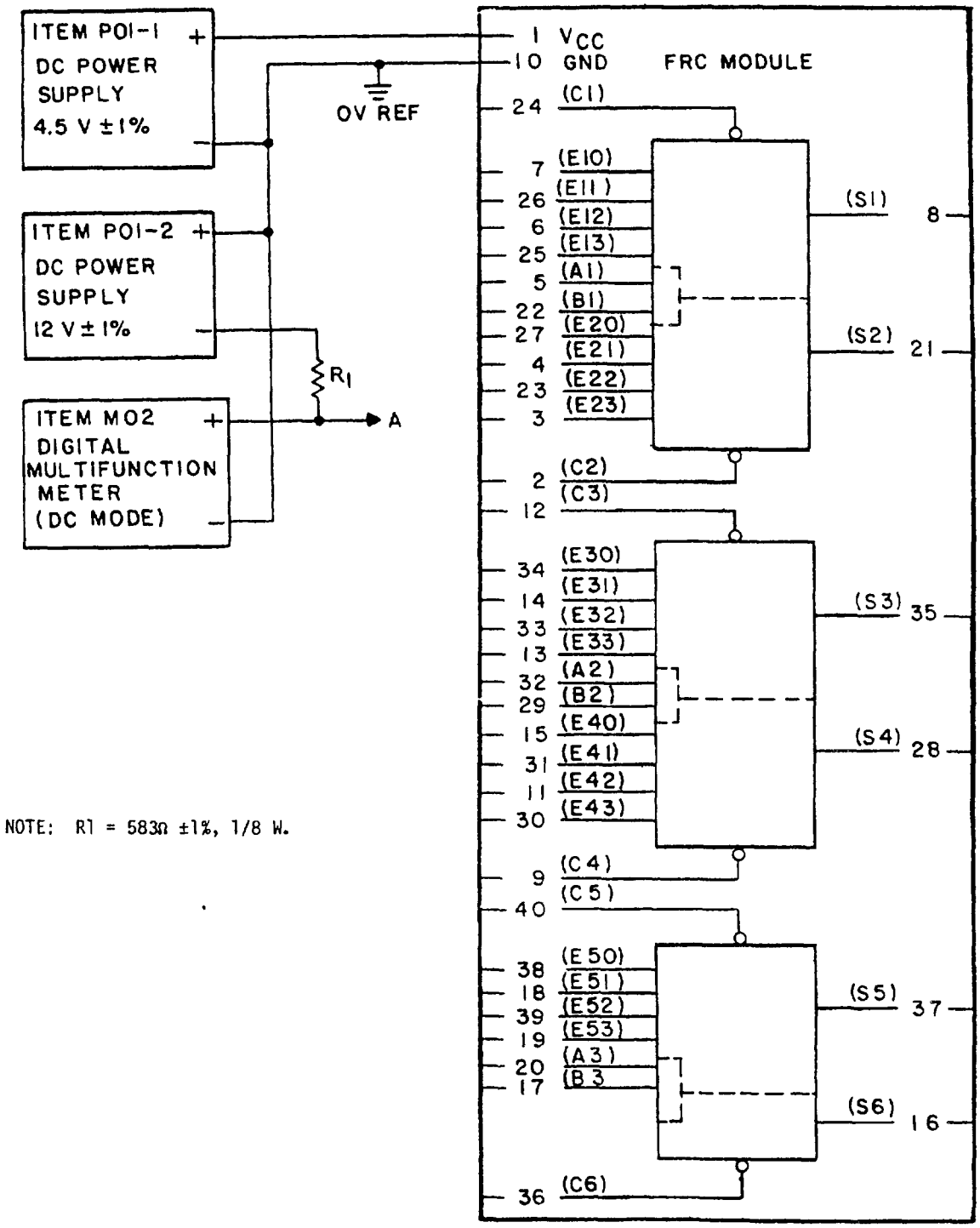
TABLE IX. Input clamp diode voltage test - Continued.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| G19 | B2 | Input clamp diode voltage | 29 | -0.4 | -1.5 | V | 12 |
| G20 | E40 | " | 15 | " | " | " | " |
| G21 | E41 | " | 31 | " | " | " | " |
| G22 | E42 | " | 11 | " | " | " | " |
| G23 | E43 | " | 30 | " | " | " | " |
| G24 | C4 | " | 9 | " | " | " | " |
| G25 | C5 | " | 40 | " | " | " | " |
| G26 | E50 | " | 38 | " | " | " | " |
| G27 | E51 | " | 18 | " | " | " | " |
| G28 | E52 | " | 39 | " | " | " | " |
| G29 | E53 | " | 19 | " | " | " | " |
| G30 | A3 | " | 20 | " | " | " | " |
| G31 | B3 | " | 17 | " | " | " | " |
| G32 | C6 | " | 36 | " | " | " | " |





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NOTE: R1 = 583Ω ±1%, 1/8 W.

FIGURE 12. Input clamp diode voltage test circuit for table IX.



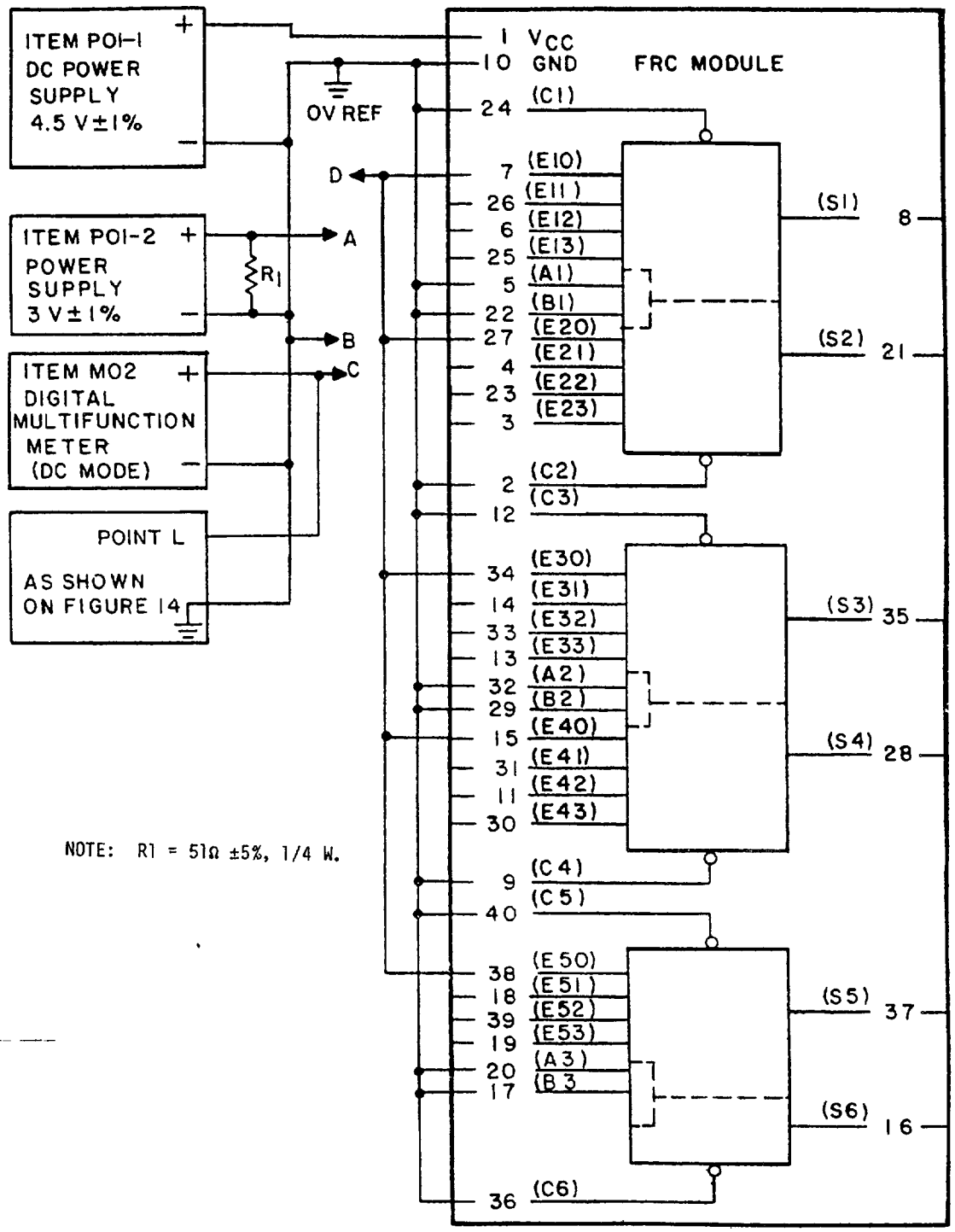
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TABLE X. Output voltage (low-level and high-level) test.

| Test condition | Parameter | | Connect | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| H1 | S1 | Output voltage, low level | Open | D | 8 | 0 | 400 | mV | 13 |
| H2 | S2 | " | " | " | 21 | " | " | " | " |
| H3 | S3 | " | " | " | 35 | " | " | " | " |
| H4 | S4 | " | " | " | 28 | " | " | " | " |
| H5 | S5 | " | " | " | 37 | " | " | " | " |
| H6 | S6 | " | " | " | 16 | " | " | " | " |
| J1 | S1 | Output voltage, high level | D | Open | 8 | 2.4 | 4.5 | V | " |
| J2 | S2 | " | " | " | 21 | " | " | " | " |
| J3 | S3 | " | " | " | 35 | " | " | " | " |
| J4 | S4 | " | " | " | 28 | " | " | " | " |
| J5 | S5 | " | " | " | 37 | " | " | " | " |



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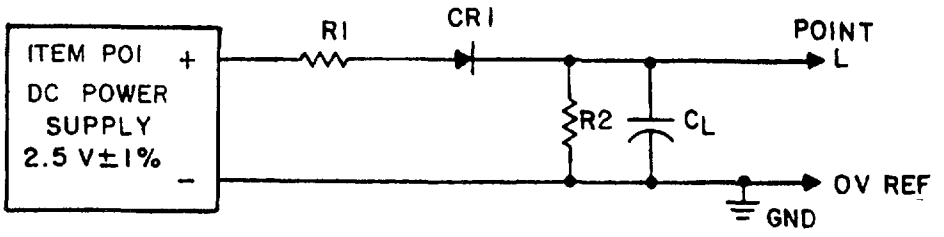



NOTE: R1 = 51Ω ±5%, 1/4 W.

FIGURE 13. Output voltage (low-level and high-level) test circuit for table X.



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NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select value for a current flow of 4 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 2.43 kΩ ±1%, 1/8 W.
4. $C_L$ = 50 ±5 pF (including noise, probe, and fixture capacitance) for test conditions N1-N41, P1-P41, Q1-Q6, and R1-R6.
   $C_L$ = 150 ±5 pF (including noise, probe, and fixture capacitance) for test conditions N42-N71 and P42-P71.
5. Capacitance $C_L$ is not required for test conditions H1-H6 and J1-J5.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 14. FRC logic load circuit.



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TABLE XI. Input current (high-level) test.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| K1 | C1 | Input current, high level | 24 | -5 | 20 | µA | 15 |
| K2 | E10 | " | 7 | " | " | " | " |
| K3 | E11 | " | 26 | " | " | " | " |
| K4 | E12 | " | 6 | " | " | " | " |
| K5 | E13 | " | 25 | " | " | " | " |
| K6 | A1 | " | 5 | " | " | " | " |
| K7 | B1 | " | 22 | " | " | " | " |
| K8 | E20 | " | 27 | " | " | " | " |
| K9 | E21 | " | 4 | " | " | " | " |
| K10 | E22 | " | 23 | " | " | " | " |
| K11 | E23 | " | 3 | " | " | " | " |
| K12 | C2 | " | 2 | " | " | " | " |
| K13 | C3 | " | 12 | " | " | " | " |
| K14 | E30 | " | 34 | " | " | " | " |
| K15 | E31 | " | 14 | " | " | " | " |
| K16 | E32 | " | 33 | " | " | " | " |
| K17 | E33 | " | 13 | " | " | " | " |
| K18 | A2 | " | 32 | " | " | " | " |
| K19 | B2 | " | 29 | " | " | " | " |
| K20 | E40 | " | 15 | " | " | " | " |
| K21 | E41 | " | 31 | " | " | " | " |





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TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| K22 | E42 | Input current, high level | 11 | -5 | 20 | µA | 15 |
| K23 | E43 | " | 30 | " | " | " | " |
| K24 | C4 | " | 9 | " | " | " | " |
| K25 | C5 | " | 40 | " | " | " | " |
| K26 | E50 | " | 38 | " | " | " | " |
| K27 | E51 | " | 18 | " | " | " | " |
| K28 | E52 | " | 39 | " | " | " | " |
| K29 | E53 | " | 19 | " | " | " | " |
| K30 | A3 | " | 20 | " | " | " | " |
| K31 | B3 | " | 17 | " | " | " | " |
| K32 | C6 | " | 36 | " | " | " | " |





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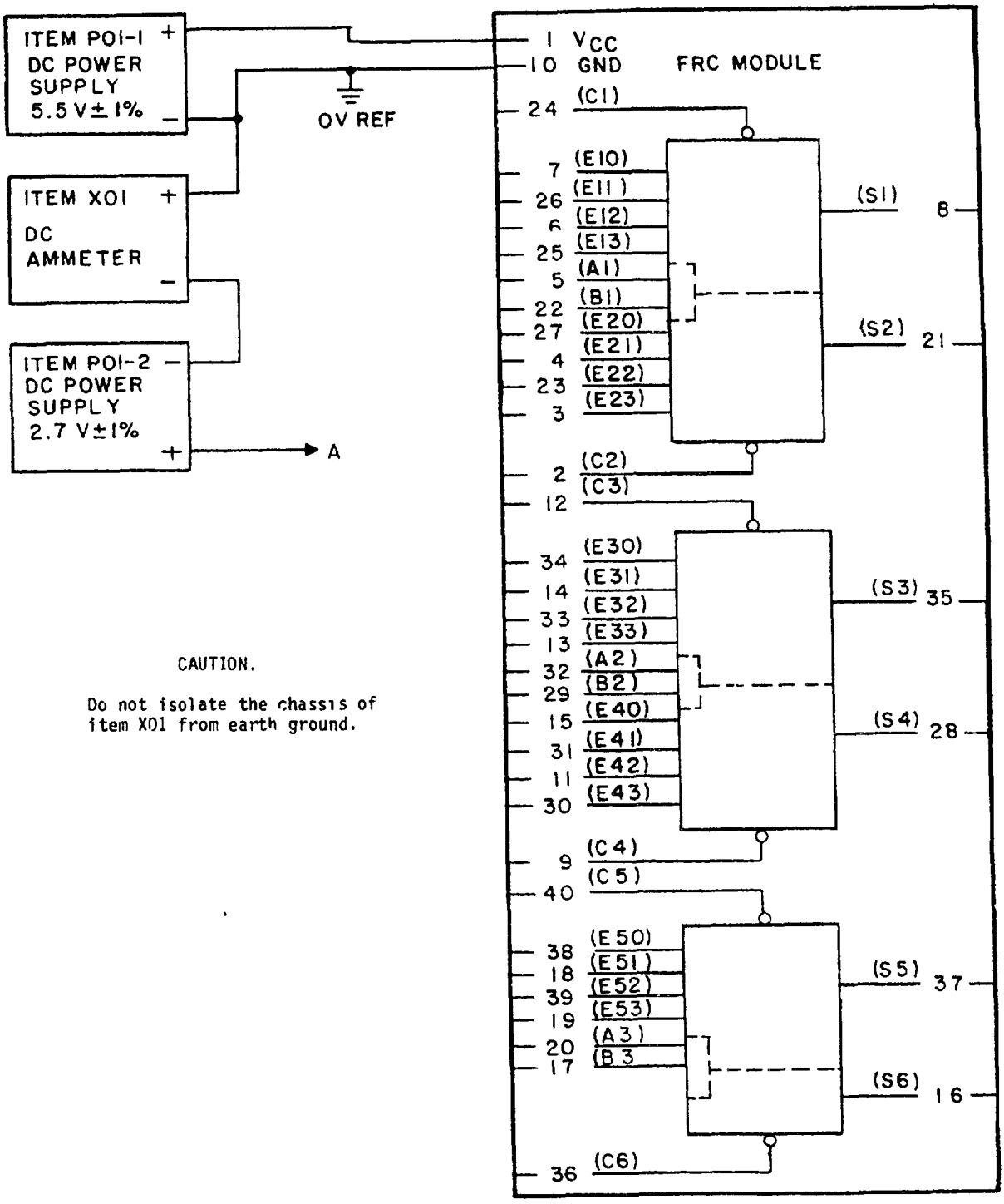



FIGURE 15. Input current (high-level) test circuit for table XI



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TABLE XII. Input current (low-level) test.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| L1 | C1 | Input current, low level | 24 | 0 | -430 | µA | 16 |
| L2 | E10 | " | 7 | " | " | " | " |
| L3 | E11 | " | 26 | " | " | " | " |
| L4 | E12 | " | 6 | " | " | " | " |
| L5 | E13 | " | 25 | " | " | " | " |
| L6 | A1 | " | 5 | " | " | " | " |
| L7 | B1 | " | 22 | " | " | " | " |
| L8 | E20 | " | 27 | " | " | " | " |
| L9 | E21 | " | 4 | " | " | " | " |
| L10 | E22 | " | 23 | " | " | " | " |
| L11 | E23 | " | 3 | " | " | " | " |
| L12 | C2 | " | 2 | " | " | " | " |
| L13 | C3 | " | 12 | " | " | " | " |
| L14 | E30 | " | 34 | " | " | " | " |
| L15 | E31 | " | 14 | " | " | " | " |
| L16 | E32 | " | 33 | " | " | " | " |
| L17 | E33 | " | 13 | " | " | " | " |
| L18 | A2 | " | 32 | " | " | " | " |
| L19 | B2 | " | 29 | " | " | " | " |
| L20 | E40 | " | 15 | " | " | " | " |
| L21 | E41 | " | 31 | " | " | " | " |





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TABLE XII. Input current (low-level) test - Continued.

| Test condition | Parameter | | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| L22 | E42 | Input current, low level | 11 | 0 | -430 | µA | 16 |
| L23 | E43 | " | 30 | " | " | " | " |
| L24 | C4 | " | 9 | " | " | " | " |
| L25 | C5 | " | 40 | " | " | " | " |
| L26 | E50 | " | 38 | " | " | " | " |
| L27 | E51 | " | 18 | " | " | " | " |
| L28 | E52 | " | 39 | " | " | " | " |
| L29 | E53 | " | 19 | " | " | " | " |
| L30 | A3 | " | 20 | " | " | " | " |
| L31 | B3 | " | 17 | " | " | " | " |
| L32 | C6 | " | 36 | " | " | " | " |



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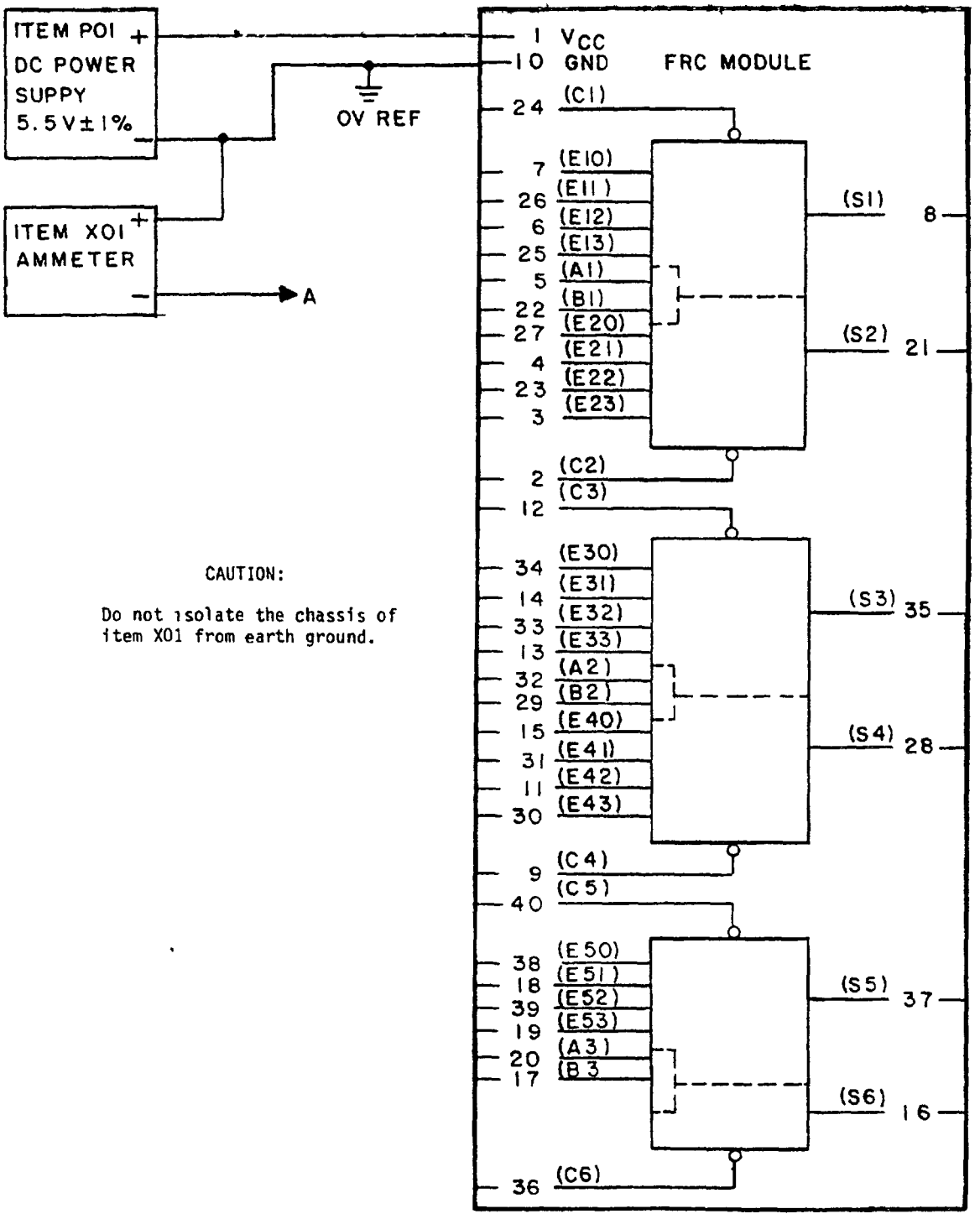



FIGURE 16. Input current (low-level) test circuit for table XII



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TABLE XIII. Power supply current (high-level, low-level and high impedance output) test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| M1 | Power supply current, high level output | C | D | 15 | 36 | mA | 17 |
| M2 | Power supply current, low level output | C, D | Open | 15 | 36 | mA | 17 |
| M3 | Power supply current, high impedance output | Open | C, D | 15 | 42 | mA | 17 |





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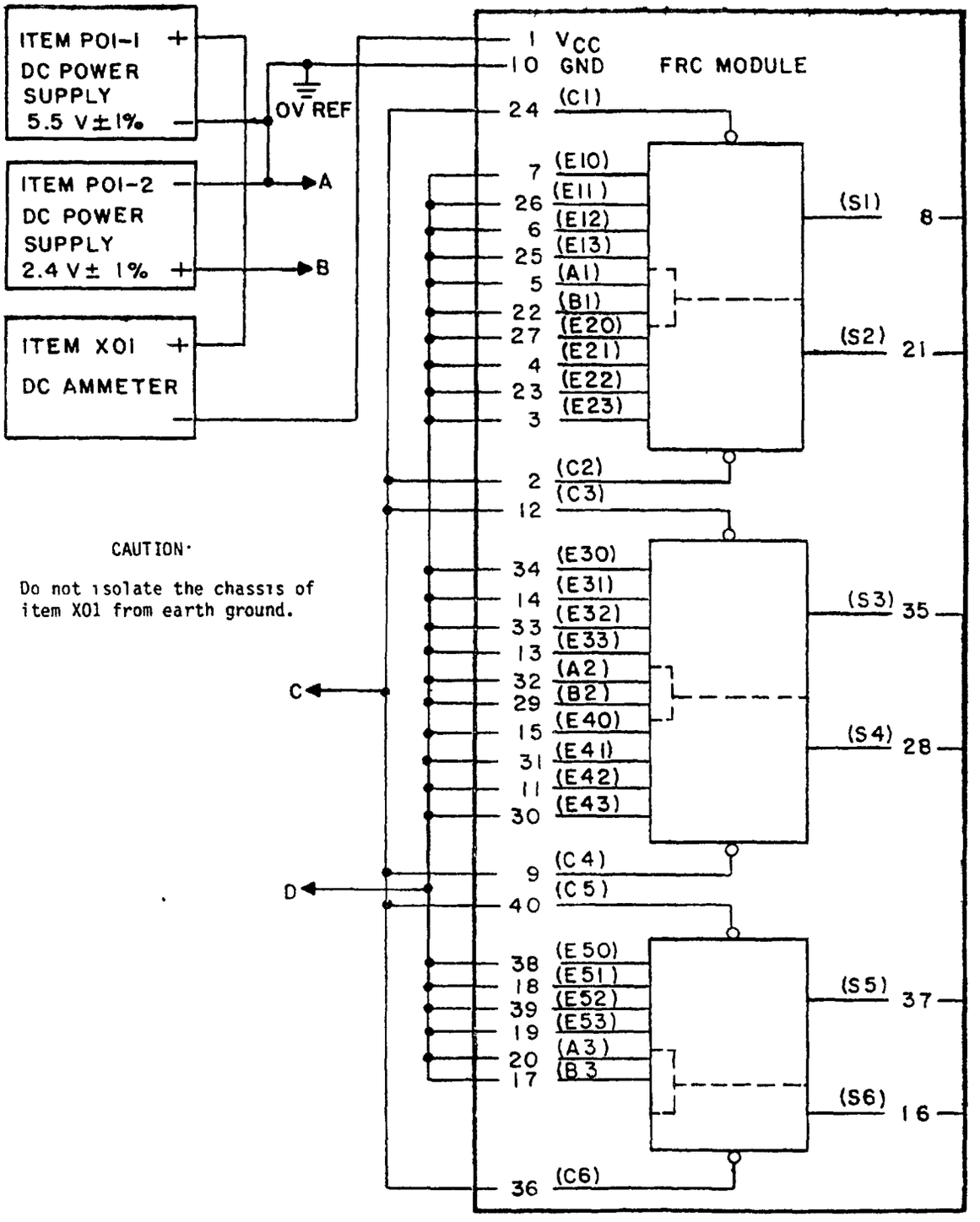



FIGURE 17. Power supply current (high-level, low-level, and high impedance output) test circuit for table XIII.



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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| N1 | E10 to S1 Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | Open | 24, 5, 22 | 7 | 8 | 5 | 29 | ns | 18,19 |
| N2 | E11 to S1 " | 22 | 24, 5 | 26 | " | " | " | " | " |
| N3 | E12 to S1 " | 5 | 24, 22 | 6 | " | " | " | " | " |
| N4 | E13 to S1 " | 5, 22 | 24 | 25 | " | " | " | " | " |
| N5 | E20 to S2 " | Open | 2, 5, 22 | 27 | 21 | " | " | " | " |
| N6 | E21 to S2 " | 22 | 2, 5 | 4 | " | " | " | " | " |
| N7 | E22 to S2 " | 5 | 2, 22 | 23 | " | " | " | " | " |
| N8 | E23 to S2 " | 5, 22 | 2 | 3 | " | " | " | " | " |
| N9 | E30 to S3 " | Open | 12, 32, 29 | 34 | 35 | " | " | " | " |
| N10 | E31 to S3 " | 29 | 12, 32 | 14 | " | " | " | " | " |
| N11 | E32 to S3 " | 32 | 12, 29 | 33 | " | " | " | " | " |
| N12 | E33 to S3 " | 32, 29 | 12 | 13 | " | " | " | " | " |
| N13 | E40 to S4 " | Open | 9, 32, 29 | 15 | 28 | " | " | " | " |
| N14 | E41 to S4 " | 29 | 9, 32 | 31 | " | " | " | " | " |
| N15 | E42 to S4 " | 32 | 9, 29 | 11 | " | " | " | " | " |
| N16 | E43 to S4 " | 32, 29 | 9 | 30 | " | " | " | " | " |
| N17 | E50 to S5 " | Open | 40, 20, 17 | 38 | 37 | " | " | " | " |
| N18 | E51 to S5 " | 17 | 40, 20 | 18 | " | " | " | " | " |
| N19 | E52 to S5 " | 20 | 40, 17 | 39 | " | " | " | " | " |
| N20 | E53 to S5 " | 20, 17 | 40 | 19 | " | " | " | " | " |
| N21 | A1 to S1 " | 6 | 24, 7, 22 | 5 | 8 | " | 49 | " | " |
| N22 | B1 to S1 " | 26 | 24, 5, 7 | 22 | 8 | " | 49 | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| N23 | A1 to S2 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 23 | 2, 22, 27 | 5 | 21 | 5 | 49 | ns | 18,19 |
| N24 | B1 to S2 | " | 4 | 2, 5, 27 | 22 | 21 | " | " | " | " |
| N25 | A2 to S3 | " | 33 | 12, 34, 29 | 32 | 35 | " | " | " | " |
| N26 | B2 to S3 | " | 14 | 12, 34, 32 | 29 | 35 | " | " | " | " |
| N27 | A2 to S4 | " | 11 | 9, 29, 15 | 32 | 28 | " | " | " | " |
| N28 | B2 to S4 | " | 31 | 9, 32, 15 | 29 | 28 | " | " | " | " |
| N29 | A3 to S5 | " | 39 | 40, 17, 38 | 20 | 37 | " | " | " | " |
| N30 | B3 to S5 | " | 18 | 40, 20, 38 | 17 | 37 | " | " | " | " |
| N31 | C1 to S1 | Propagation delay time ($t_{PHZ}$), $C_L$ = 50 pF | 7 | 5, 22 | 24 | 8 | " | 70 | " | 18,20 |
| N32 | C2 to S2 | " | 27 | 5, 22 | 2 | 21 | " | " | " | " |
| N33 | C3 to S3 | " | 34 | 32, 29 | 12 | 35 | " | " | " | " |
| N34 | C4 to S4 | " | 15 | 32, 29 | 9 | 28 | " | " | " | " |
| N35 | C5 to S5 | " | 38 | 20, 17 | 40 | 37 | " | " | " | " |
| N36 | C1 to S1 | Propagation delay time ($t_{PLZ}$), $C_L$ = 50 pF | Open | 7, 5, 22 | 24 | 8 | " | 35 | " | " |
| N37 | C2 to S2 | " | " | 27, 5, 22 | 2 | 21 | " | " | " | " |
| N38 | C3 to S3 | " | " | 34, 32, 29 | 12 | 35 | " | " | " | " |
| N39 | C4 to S4 | " | " | 15, 32, 29 | 9 | 28 | " | " | " | " |
| N40 | C5 to S5 | " | " | 38, 20, 17 | 40 | 37 | " | " | " | " |
| N41 | C6 to S6 | " | " | 20, 17 | 36 | 16 | " | " | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| N42 | E10 to S1 Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | Open | 24, 5, 22 | 7 | 8 | 5 | 39 | ns | 18,19 |
| N43 | E11 to S1 " | 22 | 24, 5 | 26 | " | " | " | " | " |
| N44 | E12 to S1 " | 5 | 24, 22 | 6 | " | " | " | " | " |
| N45 | E13 to S1 " | 5, 22 | 24 | 25 | " | " | " | " | " |
| N46 | E20 to S2 " | Open | 2, 5, 22 | 27 | 21 | " | " | " | " |
| N47 | E21 to S2 " | 22 | 2, 5 | 4 | " | " | " | " | " |
| N48 | E22 to S2 " | 5 | 2, 22 | 23 | " | " | " | " | " |
| N49 | E23 to S2 " | 5, 22 | 2 | 3 | " | " | " | " | " |
| N50 | E30 to S3 " | Open | 12, 32, 29 | 34 | 35 | " | " | " | " |
| N51 | E31 to S3 " | 29 | 12, 32 | 14 | " | " | " | " | " |
| N52 | E32 to S3 " | 32 | 12, 29 | 33 | " | " | " | " | " |
| N53 | E33 to S3 " | 32, 29 | 12 | 13 | " | " | " | " | " |
| N54 | E40 to S4 " | Open | 9, 32, 29 | 15 | 28 | " | " | " | " |
| N55 | E41 to S4 " | 29 | 9, 32 | 31 | " | " | " | " | " |
| N56 | E42 to S4 " | 32 | 9, 29 | 11 | " | " | " | " | " |
| N57 | E43 to S4 " | 32, 29 | 9 | 30 | " | " | " | " | " |
| N58 | E50 to S5 " | Open | 40, 20, 17 | 38 | 37 | " | " | " | " |
| N59 | E51 to S5 " | 17 | 40, 20 | 18 | " | " | " | " | " |
| N60 | E52 to S5 " | 20 | 40, 17 | 39 | " | " | " | " | " |
| N61 | E53 to S5 " | 20, 17 | 40 | 19 | " | " | " | " | " |
| N62 | A1 to S1 " | 6 | 24, 7, 22 | 5 | 8 | " | 59 | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| N63 | B1 to S1 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 26 | 24, 5, 7 | 22 | 8 | 5 | 59 | ns | 18,19 |
| N64 | A1 to S2 | " | 23 | 2, 22, 27 | 5 | 21 | " | " | " | " |
| N65 | B1 to S2 | " | 4 | 2, 5, 27 | 22 | 21 | " | " | " | " |
| N66 | A2 to S3 | " | 33 | 12, 34, 29 | 32 | 35 | " | " | " | " |
| N67 | B2 to S3 | " | 14 | 12, 34, 32 | 29 | 35 | " | " | " | " |
| N68 | A2 to S4 | " | 11 | 9, 29, 15 | 32 | 28 | " | " | " | " |
| N69 | B2 to S4 | " | 31 | 9, 32, 15 | 29 | 28 | " | " | " | " |
| N70 | A3 to S5 | " | 39 | 40, 17, 38 | 20 | 37 | " | " | " | " |
| N71 | B3 to S5 | " | 18 | 40, 20, 38 | 17 | 37 | " | " | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| P1 | E10 to S1 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | Open | 24, 5, 22 | 7 | 8 | 5 | 24 | ns | 18,19 |
| P2 | E11 to S1 " | 22 | 24, 5 | 26 | " | " | " | " | " |
| P3 | E12 to S1 " | 5 | 24, 22 | 6 | " | " | " | " | " |
| P4 | E13 to S1 " | 5, 22 | 24 | 25 | " | " | " | " | " |
| P5 | E20 to S2 " | Open | 2, 5, 22 | 27 | 21 | " | " | " | " |
| P6 | E21 to S2 " | 22 | 2, 5 | 4 | " | " | " | " | " |
| P7 | E22 to S2 " | 5 | 2, 22 | 23 | " | " | " | " | " |
| P8 | E23 to S2 " | 5, 22 | 2 | 3 | " | " | " | " | " |
| P9 | E30 to S3 " | Open | 12, 32, 29 | 34 | 35 | " | " | " | " |
| P10 | E31 to S3 " | 29 | 12, 32 | 14 | " | " | " | " | " |
| P11 | E32 to S3 " | 32 | 12, 29 | 33 | " | " | " | " | " |
| P12 | E33 to S3 " | 32, 29 | 12 | 13 | " | " | " | " | " |
| P13 | E40 to S4 " | Open | 9, 32, 29 | 15 | 28 | " | " | " | " |
| P14 | E41 to S4 " | 29 | 9, 32 | 31 | " | " | " | " | " |
| P15 | E42 to S4 " | 32 | 9, 29 | 11 | " | " | " | " | " |
| P16 | E43 to S4 " | 32, 29 | 9 | 30 | " | " | " | " | " |
| P17 | E50 to S5 " | Open | 40, 20, 17 | 38 | 37 | " | " | " | " |
| P18 | E51 to S5 " | 17 | 40, 20 | 18 | " | " | " | " | " |
| P19 | E52 to S5 " | 20 | 40, 17 | 39 | " | " | " | " | " |
| P20 | E53 to S5 " | 20, 17 | 40 | 19 | " | " | " | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| P21 | A1 to S1 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 6 | 24, 7, 22 | 5 | 8 | 5 | 36 | ns | 18,19 |
| P22 | B1 to S1 | " | 26 | 24, 5, 7 | 22 | 8 | " | " | " | " |
| P23 | A1 to S2 | " | 23 | 2, 22, 27 | 5 | 21 | " | " | " | " |
| P24 | B1 to S2 | " | 4 | 2, 5, 27 | 22 | 21 | " | " | " | " |
| P25 | A2 to S3 | " | 33 | 12, 34, 29 | 32 | 35 | " | " | " | " |
| P26 | B2 to S3 | " | 14 | 12, 34, 32 | 29 | 35 | " | " | " | " |
| P27 | A2 to S4 | " | 11 | 9, 29, 15 | 32 | 28 | " | " | " | " |
| P28 | B2 to S4 | " | 31 | 9, 32, 15 | 29 | 28 | " | " | " | " |
| P29 | A3 to S5 | " | 39 | 40, 17, 38 | 20 | 37 | " | " | " | " |
| P30 | B3 to S5 | " | 18 | 40, 20, 38 | 17 | 37 | " | " | " | " |
| P31 | C1 to S1 | Propagation delay time ($t_{PZH}$), $C_L$ = 50 pF | 7 | 5, 22 | 24 | 8 | " | 50 | " | 18,20 |
| P32 | C2 to S2 | " | 27 | 5, 22 | 2 | 21 | " | " | " | " |
| P33 | C3 to S3 | " | 34 | 32, 29 | 12 | 35 | " | " | " | " |
| P34 | C4 to S4 | " | 15 | 32, 29 | 9 | 28 | " | " | " | " |
| P35 | C5 to S5 | " | 38 | 20, 17 | 40 | 37 | " | " | " | " |
| P36 | C1 to S1 | Propagation delay time ($t_{PZL}$), $C_L$ = 50 pF | Open | 7, 5, 22 | 24 | 8 | " | 30 | " | " |
| P37 | C2 to S2 | " | " | 27, 5, 22 | 2 | 21 | " | " | " | " |
| P38 | C3 to S3 | " | " | 34, 32, 29 | 12 | 35 | " | " | " | " |





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TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| P39 | C4 to S4 | Propagation delay time ($t_{PZL}$), $C_L$ = 50 pF | Open | 15, 32, 29 | 9 | 28 | 5 | 30 | ns | 18,20 |
| P40 | C5 to S5 | " | " | 38, 20, 17 | 40 | 37 | " | " | " | " |
| P41 | C6 to S6 | " | " | 20, 17 | 36 | 16 | " | " | " | " |
| P42 | E10 to S1 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | Open | 24, 5, 22 | 7 | 8 | " | 34 | " | 18,19 |
| P43 | E11 to S1 | " | 22 | 24, 5 | 26 | " | " | " | " | " |
| P44 | E12 to S1 | " | 5 | 24, 22 | 6 | " | " | " | " | " |
| P45 | E13 to S1 | " | 5, 22 | 24 | 25 | " | " | " | " | " |
| P46 | E20 to S2 | " | Open | 2, 5, 22 | 27 | 21 | " | " | " | " |
| P47 | E21 to S2 | " | 22 | 2, 5 | 4 | " | " | " | " | " |
| P48 | E22 to S2 | " | 5 | 2, 22 | 23 | " | " | " | " | " |
| P49 | E23 to S2 | " | 5, 22 | 2 | 3 | " | " | " | " | " |
| P50 | E30 to S3 | " | Open | 12, 32, 29 | 34 | 35 | " | " | " | " |
| P51 | E31 to S3 | " | 29 | 12, 32 | 14 | " | " | " | " | " |
| P52 | E32 to S3 | " | 32 | 12, 29 | 33 | " | " | " | " | " |
| P53 | E33 to S3 | " | 32, 29 | 12 | 13 | " | " | " | " | " |
| P54 | E40 to S4 | " | Open | 9, 32, 29 | 15 | 28 | " | " | " | " |
| P55 | E41 to S4 | " | 29 | 9, 32 | 31 | " | " | " | " | " |
| P56 | E42 to S4 | " | 32 | 9, 29 | 11 | " | " | " | " | " |
| P57 | E43 to S4 | " | 32, 29 | 9 | 30 | " | " | " | " | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:40Z
Check the source to verify that this is the current version before use.

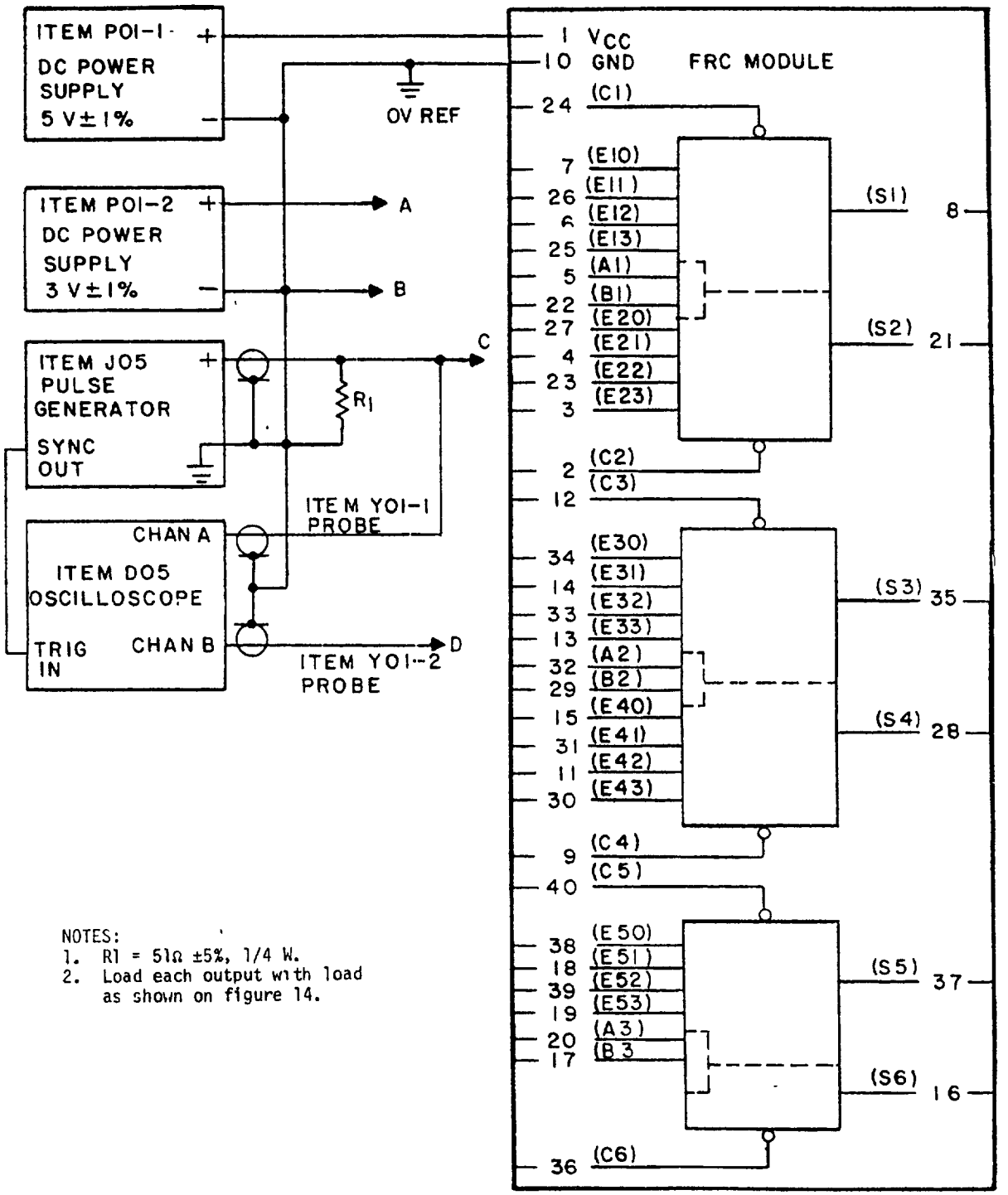
TABLE XIV. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$ and $t_{PZL}$) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| P58 | E50 to S5 Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | Open | 40, 20, 17 | 38 | 37 | 5 | 34 | ns | 18,19 |
| P59 | E51 to S5 " | 17 | 40, 20 | 18 | " | " | " | " | " |
| P60 | E52 to S5 " | 20 | 40, 17 | 39 | " | " | " | " | " |
| P61 | E53 to S5 " | 20, 17 | 40 | 19 | " | " | " | " | " |
| P62 | A1 to S1 " | 6 | 24, 7, 22 | 5 | 8 | " | 46 | " | " |
| P63 | B1 to S1 " | 26 | 24, 5, 7 | 22 | 8 | " | " | " | " |
| P64 | A1 to S2 " | 23 | 2, 22, 27 | 5 | 21 | " | " | " | " |
| P65 | B1 to S2 " | 4 | 2, 5, 27 | 22 | 21 | " | " | " | " |
| P66 | A2 to S3 " | 33 | 12, 34, 29 | 32 | 35 | " | " | " | " |
| P67 | B2 to S3 " | 14 | 12, 34, 32 | 29 | 35 | " | " | " | " |
| P68 | A2 to S4 " | 11 | 9, 29, 15 | 32 | 28 | " | " | " | " |
| P69 | B2 to S4 " | 31 | 9, 32, 15 | 29 | 28 | " | " | " | " |
| P70 | A3 to S5 " | 39 | 40, 17, 38 | 20 | 37 | " | " | " | " |
| P71 | B3 to S5 " | 18 | 40, 20, 38 | 17 | 37 | " | " | " | " |





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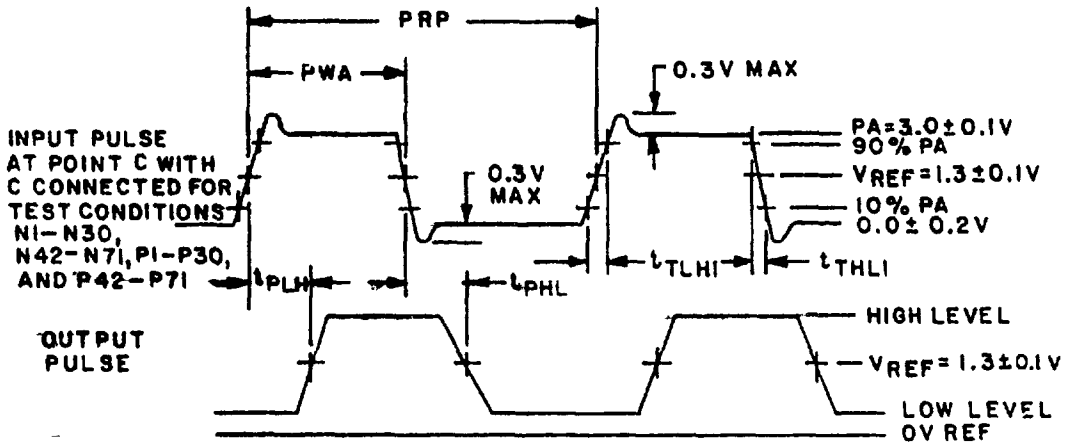



NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. Load each output with load as shown on figure 14.

FIGURE 18. Propagation delay time ($t_{PLH}$, $t_{PHL}$, $t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$, and $t_{PZL}$) test circuit for table XIV.



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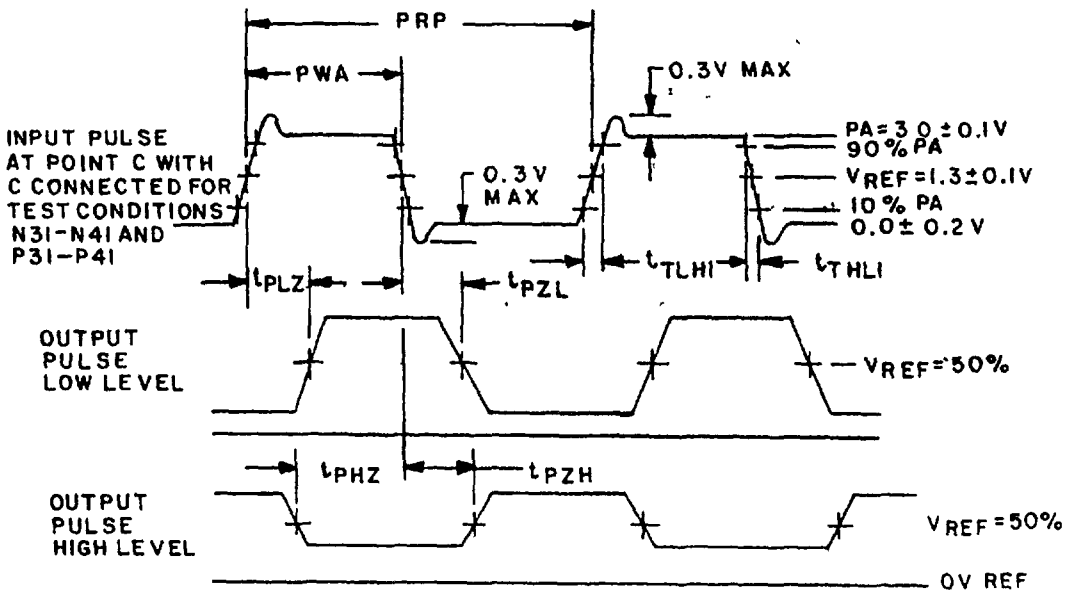



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
   PRP = 1,000 ±100 ns.
   PWA = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is >2.4 V and the low level is <0.4 V.
3. Lead length between R1 and the inputs, between items YO1-1 and YO1-2 and the outputs, and between the output pin under test to the load circuit shall be less than or equal to 1 inch (25.4 mm).
4. All input voltages are referenced to ground.

FIGURE 19. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for table XIV.



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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
   PRP = 1,000 ±100 ns.
   PWA = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is >2.4 V and the low level is <0.4 V.
3. Lead length between R1 and the inputs, between items YO1-1 and YO1-2 and the outputs, and between the output pin under test to the load circuit shall be less than or equal to 1 inch (25.4 mm).
4. All input voltages are referenced to ground.

FIGURE 20. Propagation delay time ($t_{PHZ}$, $t_{PLZ}$, $t_{PZH}$, and $t_{PZL}$) test circuit waveforms for table XIV.



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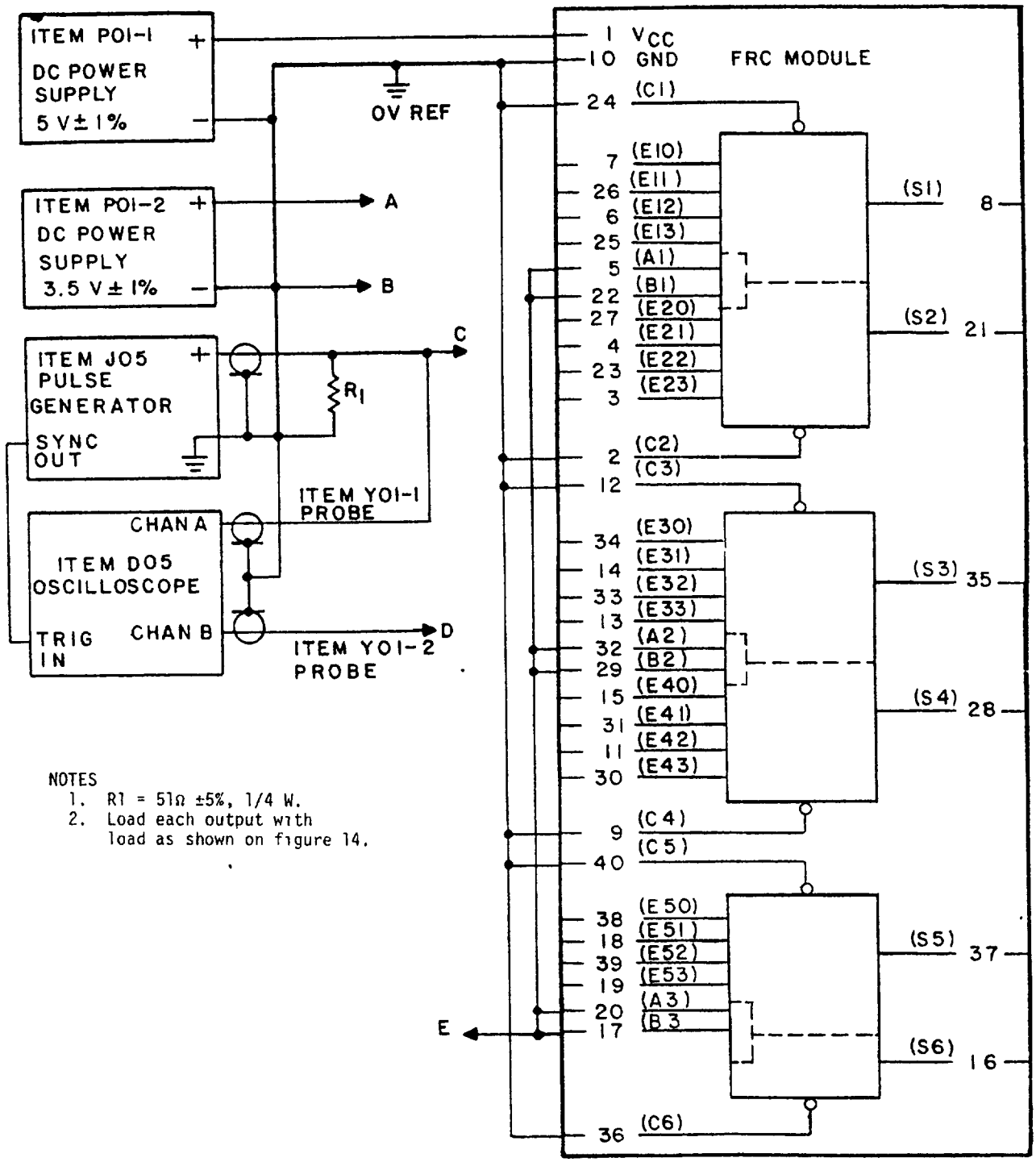
TABLE XV. Output crosstalk (low-level and high-level) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| Q1 | S1 | Output crosstalk, low level | E | 25 | 3, 13, 30, 19 | 8 | 0 | 500 | mV | 21,22 |
| Q2 | S2 | " | " | 3 | 25, 13, 30, 19 | 21 | " | " | " | " |
| Q3 | S3 | " | " | 13 | 3, 25, 30, 19 | 35 | " | " | " | " |
| Q4 | S4 | " | " | 30 | 13, 3, 25, 19 | 28 | " | " | " | " |
| Q5 | S5 | " | " | 19 | 13, 3, 25, 30 | 37 | " | " | " | " |
| Q6 | S6 | " | " | Open | 13, 3, 25, 19, 30 | 16 | " | " | " | " |
| R1 | S1 | Output crosstalk, high level | E, 25 | " | 3, 13, 30, 19 | 8 | " | 400 | mV | " |
| R2 | S2 | " | E, 3 | " | 25, 13, 30, 19 | 21 | " | " | " | " |
| R3 | S3 | " | E, 13 | " | 3, 25, 30, 19 | 35 | " | " | " | " |
| R4 | S4 | " | E, 30 | " | 13, 3, 25, 19 | 28 | " | " | " | " |
| R5 | S5 | " | E, 19 | " | 13, 3, 25, 30 | 37 | " | " | " | " |





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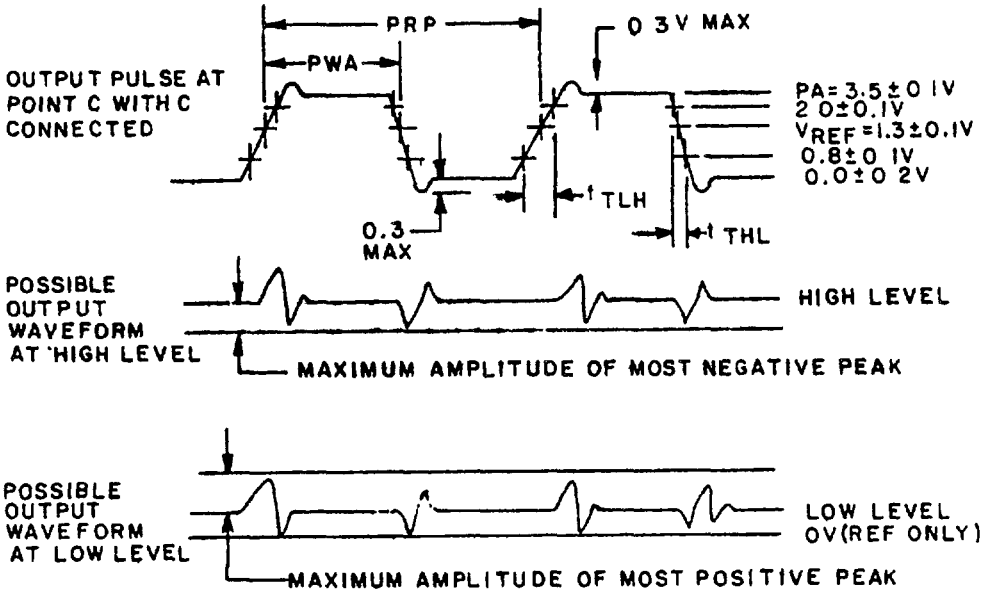



NOTES
1. R1 = 51Ω ±5%, 1/4 W.
2. Load each output with load as shown on figure 14.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit for table XV.



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NOTES
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
   PWA = 500 ns.
   PRP = 1 µs.
2. Lead length between R1 and the inputs, between items YO1-1 and YO1-2 and the outputs, and between the output pin under test to the load circuit shall be less than or equal to 1 inch (25.4 mm).

FIGURE 22. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.



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TABLE XVI. Power supply transient amplitude test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| S | Power supply transient amplitude | 0 | 400 | mV | 23, 24 |

TABLE XVII. Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| T | Power supply filter capacitance | Open | 1 | 1.6 | 3.0 | µF | 25 |



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TABLE XVIII. Input capacitance test.

| Test condition | Parameter | | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| U1 | C1 | Input capacitance | 24 | Open | 2 | 15 | pF | 25 |
| U2 | C2 | " | 2 | " | " | " | " | " |
| U3 | C3 | " | 12 | " | " | " | " | " |
| U4 | C4 | " | 9 | " | " | " | " | " |
| U5 | C5 | " | 40 | " | " | " | " | " |
| U6 | C6 | " | 36 | " | " | " | " | " |
| U7 | A1 | " | 5 | " | " | " | " | " |
| U8 | B1 | " | 22 | " | " | " | " | " |
| U9 | A2 | " | 32 | " | " | " | " | " |
| U10 | B2 | " | 29 | " | " | " | " | " |
| U11 | A3 | " | 20 | " | " | " | " | " |
| U12 | B3 | " | 17 | " | " | " | " | " |
| U13 | E10 | " | 7 | " | " | " | " | " |
| U14 | E11 | " | 26 | " | " | " | " | " |
| U15 | E12 | " | 6 | " | " | " | " | " |
| U16 | E13 | " | 25 | " | " | " | " | " |
| U17 | E20 | " | 27 | " | " | " | " | " |
| U18 | E21 | " | 4 | " | " | " | " | " |
| U19 | E22 | " | 23 | " | " | " | " | " |
| U20 | E23 | " | 3 | " | " | " | " | " |
| U21 | E30 | " | 34 | " | " | " | " | " |





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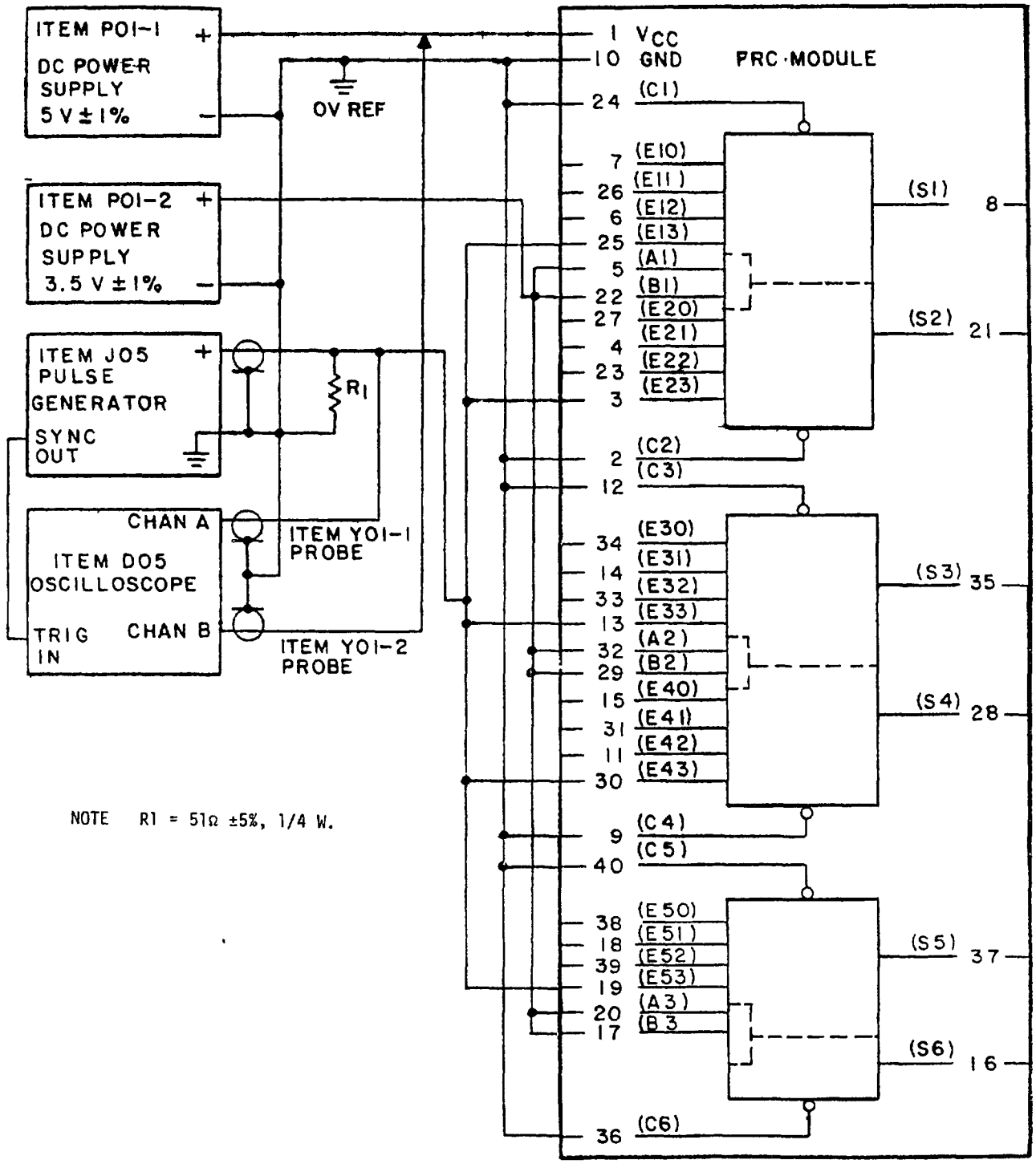
TABLE XVIII. Input capacitance test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| U22 | E31 | Input capacitance | 14 | Open | 2 | 15 | pF | 25 |
| U23 | E32 | " | 33 | " | " | " | " | " |
| U24 | E33 | " | 13 | " | " | " | " | " |
| U25 | E40 | " | 15 | " | " | " | " | " |
| U26 | E41 | " | 31 | " | " | " | " | " |
| U27 | E42 | " | 11 | " | " | " | " | " |
| U28 | E43 | " | 30 | " | " | " | " | " |
| U29 | E50 | " | 38 | " | " | " | " | " |
| U30 | E51 | " | 18 | " | " | " | " | " |
| U31 | E52 | " | 39 | " | " | " | " | " |
| U32 | E53 | " | 19 | " | " | " | " | " |





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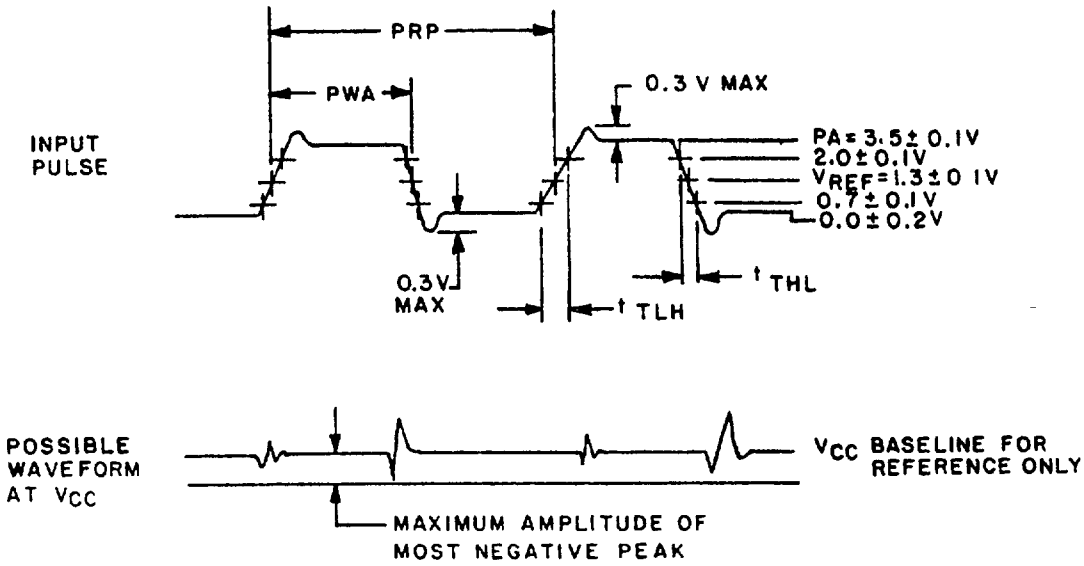



NOTE R1 = 51Ω ±5%, 1/4 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVI



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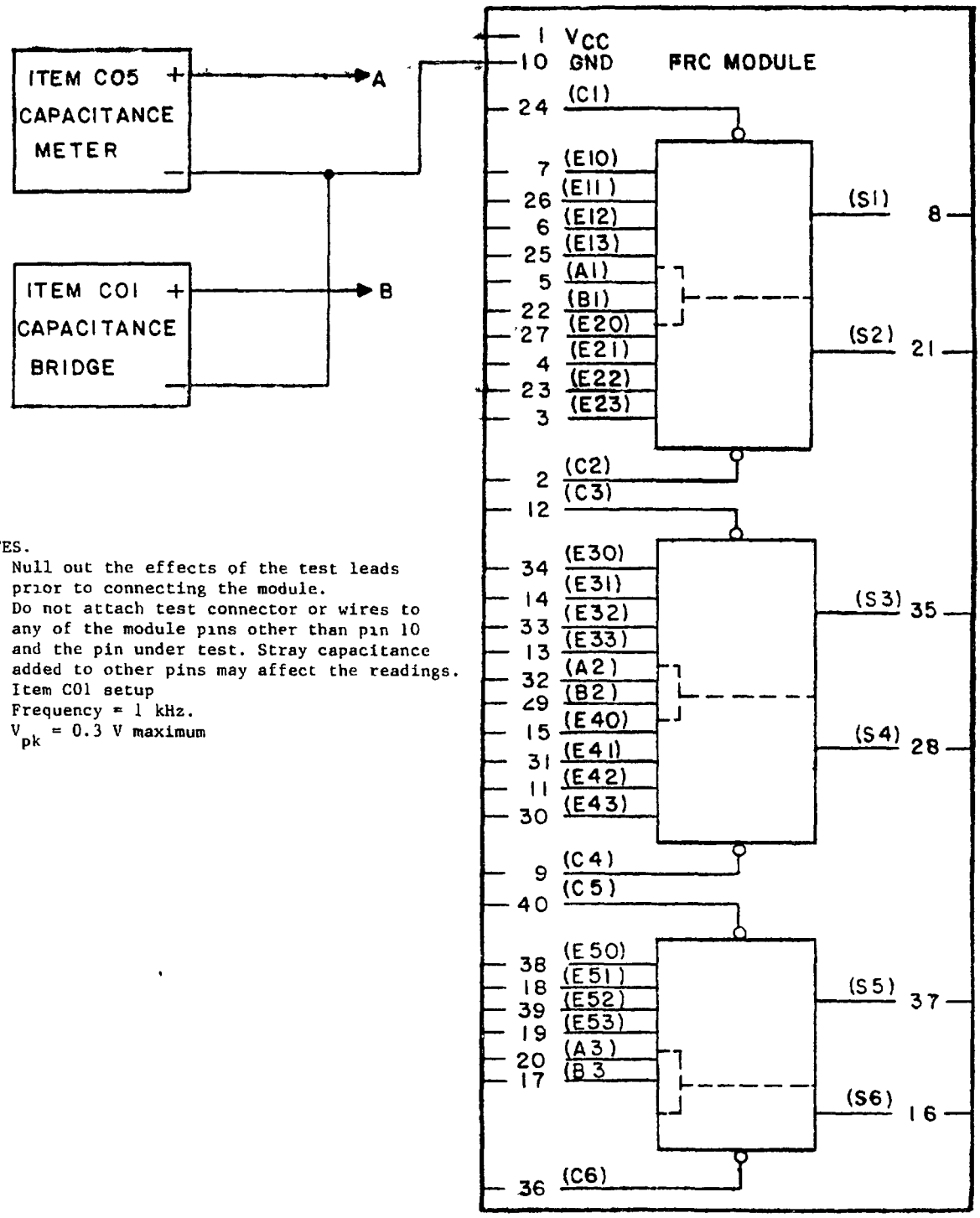



NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 6 \pm 2$ ns.
PWA = 500 ±50 ns.
PRP = 1 µs ±100 ns.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVI.



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NOTES.
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3 Item C01 setup
Frequency = 1 kHz.
$V_{pk}$ = 0.3 V maximum

FIGURE 25. Power supply filter capacitance and input capacitance test circuit for tables XVII and XVIII.

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